Ashburner Francis

AF Job No:
25-0437

Document Issue:
Rev B

Date:
04/02/2026

Architect:
Archipelago

Skygate Play – Brisbane airport

The Circuit, Brisbane Airport

D&C Specification Brief

Hydraulic Services

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Revision History

Issue	Date	Description	Initials
A	18/12/2025	D&C Pricing Issue	PMC
B	04/02/2026	D&C pricing issue Added Gameover tenancy 2	PMC

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1. PRELIMINARIES

1.1 Conditions of Contract

This section of the specification forms part of the contract documents and Principal's Project Requirements (PPR). Relevant sections of the main contract documents & PPR shall be read in conjunction with this part of the specification.

This is a **"Design & Construct" (D&C) project**. The PPR reference design provides design intent and minimum requirements only.

This section of the specification forms part of the contract documents. Relevant sections of the main contract documents shall be read in conjunction with this part of the specification.

1.2 Site Visit

Visit the site to fully evaluate the extent of work and the conditions prevailing. It is deemed essential that the Contractor visits the site to thoroughly understand the nature of the work and the new proposed and existing conditions under which the work will be executed prior to submission of the tender price.

The Contractor shall be responsible for obtaining his own information on all matters, which can in any way influence the Contract works. No claim for additional charges will be accepted for failure to comply with this requirement.

1.3 Documentation Clarification

Should there be conflicting information in the specification or drawings, the Tender shall allow for the most expensive alternative in their Tender Submission, unless written clarification in the form of an Addendum is obtained. The Tender shall obtain clarification before proceeding with the work.

2. SCOPE

2.1 General

This performance document outlines the Hydraulic services associated with this Project.

It is based on the drawings and related documents produced as part of the D&C design phase. And as such it reflects the “D & C Tender” stage of documentation. The hydraulic sub-contractor shall make all allowances necessary to price the project, based on a Design & Construct methodology and style of project delivery.

The intent of this document and any associated documents have been produced on the premise that they provide descriptions and information of sufficient information to enable a full evaluation of the complete project to be gained by:

- Using the descriptions and details provided.
- Relying on the tender’s expertise to make the appropriate allowances by extrapolations, inference, deduction, assumption and the like to ensure a complete and operational hydraulic system is provided that meets and satisfies the clients expectations and operational requirements.
- Where items are not specifically shown on the drawings or included within this document but are required for the type of facility proposed, taking the quality and extent of the provision from the standards set by the information provided in the documentation and including for all such requirements.

This document is NOT intended to be an exhaustive brief for the project. It is intended to:

- Set a minimum standard for the quality both in materials and workmanship for all hydraulic works and systems.
- Be a basis for the production of documentation to be used to lodge for Plumbing Approval with Local Council, QFD / ARFF, BAC approvals and For Construction Documentation.

The hydraulic drawings and specification indicate the concept design requirements for this project and set about the scope to allow competitive tenders to be called.

The contactor shall be responsible for the completion of design and detailed services co-ordination and all design drawings shall be submitted to the principal’s representative by the contractor for final approval, prior to lodgement with all required authorities.

All work shall be carried out in accordance with the specification, drawings and such details and instructions as are issued by the principal’s representative during the progress of the work.

This specification shall be read in conjunction with the drawings, all of which are intended to be mutually explanatory. All work called for by the one even if not by the other, shall be fully executed.

The drawings and specification show the general intent and do not show all equipment required to complete the hydraulic works. Items as used in normally accepted good trade practice with installations of this type and which are not specifically mentioned, shall be included and allowed within the tender pricing and installation.

All works shall be carried out in accordance with all BAC standards, guidelines and typical drawings; The Contactor shall be familiar with the extent of work required under BAC projects and allow for all specific equipment and works accordingly.

The work carried out shall comprise the manufacture, supply, delivery, installation, safe work practices and procedures, painting, testing, commissioning, shop drawings, set-out drawings, as-built drawings, block plans, maintenance manuals, defects liability, training, maintenance, guarantee and service under warranty of the Hydraulic Services for this project.

The work includes the provision of all supervision, materials, labour, freight, cartage, tools, plant, appliances and other work necessary, whether specifically mentioned or not.

All work shall be carried out in accordance with the Specification, Drawings, manufactures recommendations and industry best practice and such details and instructions as are issued by the Superintendent or their representative during the progress of the work. The equipment installed shall be handed over to the client in a fully operational state as intended for the equipment function and use.

This Specification shall be read in conjunction with the Drawings, all of which are intended to be mutually explanatory. All work called for by the one even if not by the other, shall be fully executed. Where ambiguities or discrepancies occur in the documents, the most onerous of them shall be allowed for, except where clarification is sought prior to contract award.

The Drawings and Specification show the general intent and do not necessarily show all equipment required to complete the hydraulic works. Minor items as used in normally accepted trade practice with installations of this type and which are not specifically mentioned, shall be included in the Tender and installation.

2.2 Project Description

The hydraulic drawings supplied are diagrammatic and approximate only.

These drawings, together with this performance document and all associated consultant's drawings and briefs are intended to be mutually explanatory and to show the scope of the work required for this D&C Tender Stage. All work set forth by one, if not by the other, shall be fully executed.

The work covered within the drawings and documents produced by the hydraulic services sub-contractor shall comprise the finalisation of design, co-ordination, documentation, safety in design and construction, supply, installation, testing, commissioning, quality assurance, certification, and maintenance during the defects liability period for the provision of new hydraulic systems as indicated within this document and the following drawings to serve the development.

The Hydraulic services sub-contractor shall be responsible for the detailed design activities listed below in addition to those activities normally undertaken during construction and shall achieve occupational approval for the overall facility.

The detailed design shall be fully co-ordinated and compatible with the remainder of the project design.

The Ashburner Francis DWG drawings used to produce D&C documentation will not be provided to the D&C Contractor as it is protected intellectual property.

Design responsibilities include, but not limited to:

- Production of client approval drawings.
- Production of co-ordination drawings.
- Production of and submission of drawings for authority approval.
- Production of For Construction drawings.
- Production of As-built drawings.
- Design drawings shall be produced in A1 size, generally at 1:100 for layouts & 1:10 to 1:50 for details.
- In addition to each approval milestone, the sub-contractor shall provide a statement certifying that design complies in all respects to all referenced & supplied documents.
- Undertake detailed spatial co-ordination of the services with respect to all other works and trades including the detailing of interfaces. The contractor is responsible for ensuring all interfaces are covered and that there no scope gaps.
- Perform & confirm final calculations for all hydraulic requirements, including Pump duties, pipe sizes & velocities, pressure loses, hotwater demands, flow rates etc.
- Design and co-ordinate the supports for the services and equipment.
- Design and select fire stopping for all services penetrations.
- Size and location of penetrations.
- Coordination of interfaces for all trades. Whether or not explicitly identified in the tender documentation, the contractor shall establish all the required interfaces between each service and coordinate the provision of such interfaces. This shall be considered when preparing tender price and undertaking final design.
- Co-ordination of installation with other trades.
- Co-ordination of the construction of the installation.
- Accommodate thermal and ground movement expansion and construction joint movement in all services taking into account final installation details and consistency with the specified requirements.
- Final selection of all valves, pipework and hydraulic plant and equipment.
- The preparation of hydraulic drawings sufficient to allow construction of hydraulic systems, including submissions where required for approvals and authority permits.
- Allow for detailed co-ordination in specific locations eg. Plant rooms, risers, corridors, ceiling & floor voids, hotwater ventilation requirements etc.
- Technical co-ordination of all the elements of the hydraulic works and any suppliers.
- Where the contractor wishes to use alternative methods or equipment, fully evaluate and submit a detailed report on any proposals for review by the Principal's Representative.

2.3 Extent of Work

The work carried out shall comprise the completion of the design drawings, manufacture, supply, delivery, installation, safe work practices and procedures, painting, testing, commissioning, block plans, maintenance manuals, defects liability, training, maintenance, guarantee and service under warranty of the Hydraulic Services for this project. The work shall also include the provision of all supervision, materials, labour, freight, cartage, tools, plant, appliances and other work necessary, whether specifically mentioned or not.

The work shall comprise the complete design, supply, installation, excavation and backfilling, coring, staging, coordination, out of hours works, testing and commissioning which includes but is not limited to the following main elements:

The work shall comprise the complete supply, installation, excavation and backfilling, coring, staging, coordination, out of hours works, testing and commissioning which includes but is not limited to the following main elements:

- Liaison with all required local authorities, including submissions where required for approvals and obtaining all authority permits and certificates to allow the progress and completion of the works and shall include, but not be limited to: Stormwater, Sewer, Water connections (by Civil), obtaining all plumbing permits, QFD / ARFF permits, BAC requirements & final sign-off.
- 4 Star Greenstar initiatives (not certification) is being targeted for this building, as such the design and construction of the electrical services shall align with the selected points table as issued by the Greenstar Professional.
- Production of hydraulic drawings suitable for Client Review, Approval, Co-ordination, Workshop and As-built drawings to BAC requirements.
- Installation to meet Tenant Brief, PPR documents & BAC requirements.
- Design and Installation certificates including Form 15 & Form 12.
- Coordination of Flow and Pressure testing with BAC, and allowance to determine exact allowable fire water demand for the site.
- Contractor shall allow to undertake additional survey of existing services prior to commencement, to confirm invert levels, sizes and locations of interface connections are correct prior to commencement of installations. Allow to make site adjustments to avoid and coordinate with existing services as required. Where services clashes, invert levels and sizes do not match information provided and where in doubt, contact the project manager, builder, architect and hydraulic consultant for advice.
- Liaison and coordination with all services on-site, including coordination with Structure, Civil, Mechanical, Fire and Electrical contractors & site adjustment of final hydraulic services to suit various shop drawings, existing services & changes made throughout the construction process, shall be allowed for.
- Supply and utilisation of safety equipment necessary to carry out the works.
- Mobile elevated work platforms and in-ground trench shoring as required.
- Greenstar requirements & metering
- Site Investigation and Supervision.
- Underground and utilities locating.
- Temporary works.
- Dewatering.
- Removal & disposal of redundant services to the satisfaction of the local authority.
- Sealing and disconnection of disused services.
- Reinstatement of building fabric and surfaces.
- Trenching and backfilling.
- Sanitary drainage & plumbing.
- Storm water & rainwater drainage.
- Condensate drainage.
- Cold water services.
- Hot & warm water services.
- Non potable water services.
- Rainwater services.
- Recycled water conduit.
- Irrigation services. (Capped water point for use by Irrigation contractor, ensure to coordinate with irrigation contactor prior to installation)

- Rainwater services.
- Rainwater storage tanks.
- Fire hydrant services.
- Fire hose reel services.
- Capped points for continuation by sprinkler trade – allow to coordinate with sprinkler trade for exact locations and sizes on-site.
- Drainage point(s) for Sprinlker trade - allow to allow to coordinate with sprinkler trade for exact locations and sizes on-site.
- Bollard protection / sheaths to pipework services to BAC requirements
- Ensure to allow for sufficient water metering to meet BAC and Greenstar requirements.
- Insulation of piping systems.
- Allow for supports suitable for seismic (earthquake) restraints to AS1170.4 where applicable.
- Identification of services
- Backflow prevention valves.
- Sanitary fixtures, fittings, and associated equipment.
- Core Holes.
- Sealing of all penetrations as required to maintain the integrity of the Fire Rated elements.
- Sealing of all penetrations as required to maintain the integrity of hygiene requirements (Pipetites).
- Allow for all commissioning and testing of the equipment, pump sets, systems, plant and installation.
- Registration of backflow prevention valves with the local authority.
- Provision of samples and technical information.
- Shop drawings, Set-out, Block plans (including Fire Hydrant, Gas & Hotwater) and As-built drawings.
- Photographic records of all installed in-ground services.
- Operational and Maintenance manuals incorporating as-built drawings including retained existing services and photographs & CCTV of installed works.
- CCTV survey & Jet cleaning of all drainage services prior to practical completion.
- On-site Training.
- Ongoing maintenance during the DLP (defects liability period).
- Re-testing and registration of all temperature control & backflow devices prior to the end of DLP.
- Fire Hydrant Block plan required to meet AS 2419 & QFD & BAC requirements

2.4 Work In Other Trades

Related work in connection with the Hydraulic Services (performed by other trades) includes, but is not limited to the following:

2.4.1 By Main Contractor/Builder

- Coordination of all trades.
- Site facilities and ablutions.
- Site temporary water and electrical services.
- Duct access panels and inspection panels.
- Supply and installation of Fire Extinguishers & Fire Blankets.
- Stainless steel bench tops incorporating sinks.
- Vanity basin bench top and hole preparation.
- Sink cupboard bench top and hole preparation.
- Temporary water and sewer drainage during construction.
- Bollard protection / sheaths to pipework services to BAC requirements
- Concrete spoon drains.
- Rebates in concrete to accept trench drain and sump frames.
- Reinstatement of concrete slabs & road services.
- Concrete slab to base of Fire Booster Assembly - Slab to fall away to prevent pooling of water

- Waterproof membranes.
- Bunding & waterproofing to water meter cupboards to suit authority requirements.
- Signage to hydraulic services equipment and plant.
- Painting of exposed pipework (where required)
- Floor grading to floor drains.
- Falls in slabs to direct rainwater away from buildings.
- Falls in external ground finishes to direct rainwater away from buildings.
- Support duct and frame walls behind fixtures and fittings complete with access panels (Note: Hydraulic subcontractor to provide or coordinate details for requirements).
- Datum set outs at each floor level.

2.4.2 By Mechanical Contractor

- All toilet and room exhaust systems.
- Ventilation to hydraulic plant rooms.
- Extension of condensate wastes from AC units to tundishes. (as provided by Hydraulic contractor)
- Control or monitoring wiring from all control panels, meters and all other items associated with the hydraulic services to the BMS and RUMS – Terminal strips to be provided by hydraulic contactor as interface.

2.4.3 By Electrical Contractor

- Temporary electrical supply for use during construction.
- Power / cabling to systems as required to all water meters. Hydraulic contractor to coordinate & provide details to electrical trade
- Supply of electrical power and connection to hot water units, dishwashers, boiling & chilled water units, electronic flushing devices, refrigerated drinking fountains, rainwater pumps, control panels and all other plant & associated equipment requiring electrical connections.
- Supply and installation of energy meters configured to enable individual time-of use energy data recording in accordance with the NCC ‘energy efficiency’ requirements ‘section J’, to all buildings with a floor area more than 2500m2.
- Alternative sources of electrical supply shall be provided to all control panels requiring a back-up electrical supply.

Note:
Allowance for wiring between pump hydraulic equipment control panels and hydraulic plant shall be undertaken by the hydraulic contractor or their appointed sub-contractor.

2.4.4 By Fire Protection Contractor

- Provision of Automatic Fire Detection and Alarm System, Automatic Sprinkler System, Emergency Warning System and Fire Extinguishers.
- Continuation of capped sprinkler water supply & drainage (as provided by Hydraulic contractor) Allow to coordinate exact locations and sizes on-site with Hydraulic Trade
- Continuation and extension of sprinkler test drains. (as provided by Hydraulic contractor)

2.4.5 By Civil contractor

- Supply and install metered and capped Water and Fire service stubs (for continuation by the Hydraulic subcontractor).
- Supply and install capped Stormwater and Sewer service stubs (for continuation by the Hydraulic subcontractor).
- Installation of subsoil drains to external landscaping retaining walls & kerbs.

2.4.6 By Roofing Contractor

- Supply and install all roofs, gutters, gutter sumps, rainwater heads.
- Provide flashings to all penetrations through the roof.
- Overflows to box gutters.
- Hail guards to BAC requirements
- Coordinate gutter pop locations with hydraulic subcontractor

2.4.7 By Landscape Contractor

- Continuation of capped water supplies (as provided by Hydraulic contractor).
- Installation of subsoil drains to retaining walls & kerbs.

2.5 Deliverables

The contractor shall make themselves familiar with and make allowances for **all BAC design & construction deliverables**, including but not limited to:

- Schematic Design
- Developed Design
- Detailed Design
- Construction
- Witness and Hold Points
- Pre-Commissioning Tests
- Commissiong
- Post-Construction and maintenance

For detailed requirements refer to the minimum following applicable documents:

- **BAC Design Guidelines (DG series of documents)**
- **BAC Airport Design Guidelines (ADG series of documents)**
- **BAC Airport Approval Guidelines (AAG series of documents)**
- **BAC Airport Construction Guidelines (ACG series of documents)**
- **BAC Environment Policy and plans**

2.6 Drawings

Refer to the issued drawings that form part of this D&C specification.

- The supplied drawings are intent drawings only and are considered to be 50% documentation only. The contractor shall take this into account when providing D&C pricing.

3. GENERAL REQUIREMENTS

3.1 General

This section outlines general requirements which are applicable to all sections of this specification and the works.

3.2 Standards and Regulations

Conform to the latest editions of all standards and regulations, except where other editions or amendments are required by statutory authorities.

The following standards and regulations are applicable:

- BAC design & construction guidelines.
- Cold Water supply: To AS/NZS 3500.1.
- Plumbing and drainage: To AS/NZS 3500.2.
- Stormwater: To AS/NZS 3500.3.
- Hot Water supply: To AS/NZS 3500.4.
- Gas: To AS/NZS 5601.
- Fire Hydrants: To AS/NZS 2419.
- Fire Hose Reels: To AS/NZS 2441.
- National Construction Code (NCC) volume 1 & 3
- Watermark Certification Scheme
- Queensland Development Code (QDC)
- Queensland Plumbing and Waste Water Code
- Queensland Plumbing & Drainage Act and Regulation
- State Workplace/Occupational Health and Safety Legislation
- All other relevant Acts and Regulations, Local Authority Requirements, Australian Standards and Codes having jurisdiction.

It is the responsibility of the Hydraulic Contractor to carry out all liaison and coordination with all required authorities and to ensure satisfaction with their requirements.

The subcontractor shall make themselves aware of and ensure that following documents are taken into consideration, including but not limited to:

- Fire Engineering Report
- Acoustic Engineering Report
- Performance Based Solution (PBS)
- Soil Survey Report
- Greenstar documentation
- ESD documentation
- Authority / BAC Water Metering Guidelines
- BAC Design Guidelines (DG series of documents)
- BAC Airport Design Guidelines (ADG series of documents)
- BAC Airport Approval Guidelines (AAG series of documents)
- BAC Airport Construction Guidelines (ACG series of documents)
- BAC Environment Policy and Plans

3.3 Authorities

The relevant Authorities having interest with this installation shall include but not be limited to the following.

Brisbane Airport Corporation	BAC
Plumbing & Drainage Authority	Brisbane City Council (BCC) & BAC
Water & Sewer Utility	Queensland Urban Utilities (QUU)
Fire Services	Queensland Fire Department (QFD) & Aviation Rescue Fire Fighting Services (ARRF)
Gas Service	Queensland Department of Resources & Mines AGIG (Australian Gas Infrastructure Group) & Gas Retailer

Responsibility: The Hydraulic contractor shall liaise with the Local Authorities for all inspections, final approvals and relevant fees where applicable.

The Hydraulic contractor shall advise the Hydraulic Consultant if requested to alter any part of the design by the Local Authority. Variations in altering the design will not be approved if the hydraulic contractor carries out any alterations without approval by the Hydraulic Consultant.

3.4 Authority Applications & Chargers

The Hydraulic subcontractor shall prepare and submit for approval all necessary documents to obtain authority approval. Allow to pay all fees and charges associated with obtaining authority approvals associated with the scope of works.

3.5 Inspections

Provide a minimum of 2 days' notice to the local authority prior to concealing any pipe or fittings for approval, including any in-ground drainage, water supply, fire services etc. Do not conceal any pipes until approval has been obtained from the local authority. Failure to comply with this clause will result in the subcontractor excavating and reinstating those parts covered at the subcontractor's expense.

3.6 Testing

Except for site tests, all tests shall be carried out by authorities accredited by NATA to test in the relevant field, or an organisation outside Australia recognised by NATA through a mutual recognition agreement. Co-operate as required with testing authorities.

Site tests shall use instruments calibrated by authorities accredited by NATA.

Submit copies of test reports, including certificates for type tests, showing the observations and results of tests and compliance or non-compliance with requirements. These shall be included within maintenance manuals.

If tests are to be carried out on parts of the works, do not conceal those parts and do not commence further work on those parts until the tests have been satisfactorily completed and compliance verified.

Maintain a separate set of drawings on site to record progress of all testing.

3.7 Meetings

The Sub-Contractor shall allow for and attend all meetings to co-ordinate with other services and trades and as requested by the Superintendent.

3.8 Hydraulic Subcontractors Submissions

Submit documents for review and approval in a timely manner, to suit the construction program. Advise if any of the documents are required to be returned. Co-ordinate submissions of all related items. Do not cause delays by making late or inadequate submissions. Issue a minimum of one paper copy of any submission made.

Submissions shall identify the project, subcontractor, subcontractor or supplier, manufacturer, applicable product, model number and options, as appropriate and include pertinent contract document references. Include service connection requirements and product certification. Identify non - compliance's, project requirements, and characteristics, which may be detrimental to successful performance of the completed work.

Do not commence work affected by subcontractor's submissions until, the submissions have been endorsed as satisfactory. If a document contains errors, submit a new or amended document as appropriate, indicating changes since the previous submission. If part or all of an installation is to be designed by the subcontractor, submit documents showing the layout and details of the installation.

If it is proposed to change the installation from that shown on the contract documents, or if statutory authorities require changes, submit variation documents showing the proposed changes prior to carrying out any changes.

3.9 Workmanship

The whole of work shall be carried out by or under the full supervision of a fully licenced Plumbing Contractor and be of a high standard throughout.

Where the principal considers workmanship is not consistent with good trade practice or standards or that any plant is inferior in quality or deficient in quantity to that specified, notwithstanding that a progress payment may have been made in respect thereof, the principal may give written notice of a defect or deficiency. Rejected work or materials shall be removed from the site within 48 hours of such rejection. Defective work shall be reconstructed to conform to the specified requirements.

3.10 Relevant Parties

The following terms shall represent the relevant parties in this specification: -

"Builder"	-	The head contractor awarded the project
"Contractor"	-	The head contractor awarded the project
"Sub-Contractor"	-	The Hydraulic Services Sub-Contractor
"Superintendent / Principle"	-	The client's representative for the construction works.

3.11 Contract Documents

Layouts of service lines, plant and equipment shown on the drawings are diagrammatic only, except where figured dimensions are provided or calculable. Before commencing work, obtain measurements and other necessary information. The sub-contractor is to verify and confirm all scales on hydraulic services drawings via site measurement and/or verified by architectural drawings to be used as a reference.

If the design drawings are to be altered in any way from that shown, the sub-contractor shall not alter any part of the design without written approval from the hydraulic consultant. Should alterations occur without prior approval, the sub-contractor may be required to remove part or all of the work installed and reinstall the works as per the design drawings. These reworks shall be completed at no additional cost.

The sub-contractor shall allow for all staging of the works where required, including: out of hours works, people and traffic control to minimise disruptions and maintain existing services where required. The sub-contractor shall also obtain a suitable close down time and / or give notification to local authorities having jurisdiction over the affected services

Invert levels on the drawings have been obtained from on-site inspections, survey information, verbal advice or local authority information and must be checked and confirmed on site before excavation or installation of pipework to ensure connection to the connection points is achievable providing correct cover and fall. Failure to confirm invert levels are correct renders the sub-contractor liable for any rectification works.

Spot levels take precedence over contour lines and ground profile lines.

3.12 Existing Services

All existing service locations are taken from existing information (where available) and existing hydraulic information is assumed only; no responsibility is taken for the accuracy of the existing services information supplied. The sub-contractor is to conduct site inspections & allow for all costs for hire or purchase of pipe locating equipment and services to ensure these services are accurately located prior to the comment of works.

3.13 Design Criteria

The hydraulic services installation shall comply with the following design criteria:

- Water services in general shall have a maximum velocity of 2.4 m/s.
- Hot water circulatory return systems shall have a maximum velocity of 0.8 m/s.
- Minimum pressure at any fixture outlet shall be 150 kPa.
- Maximum pressure at any fixture outlet shall be 500 kPa.
- Maximum temperature at any sanitary fixture used for personal hygiene shall be 50°C.
- Maximum temperature at any PWD, Child Accessible or Aged Care fixture used for personal hygiene shall be 45°C.
- Maximum temperature drop through the hot water return system shall be 5°C.
- The maximum calculated wait time for hot water to arrive at any outlet shall not exceed 20 seconds.
- Maximum Temperature drop through the warmwater return system shall be 3°C.
- The maximum length of warmwater dead legs shall not exceed 10 meters.
- The fire system shall be designed with a maximum velocity of 4m/s and maximum pressure loss of 150 kPa during QFD / ARFF boosting.

3.14 Fire Ant Zone

It is the hydraulic sub-contractor's responsibility to review and confirm if the project works are to be undertaken in a Fire Ant Zone. If the works are located within a Fire Ant Zone, the sub-contractor must notify Biosecurity Queensland and develop a suitable risk management plan prior to the commencement of works.

3.15 Manufacturer's Warranty

Provide manufacturer's warranty agreements where applicable.
Complete all necessary documentation on the Owner's behalf.
Include copies of warranties in the maintenance manuals.

3.16 Certification

Certify that the works have been installed to the requirements of all applicable rules and regulations.

Provide a Form 12 at completion of the works.

3.17 Hazard Assessment and Risk Management

Comply with the requirements of the applicable State Occupational / Workplace Health and safety Act and Regulations with respect to hazard identification and risk management.

Where required submit a hazard assessment and risk management plan for the construction phase of the works.

3.18 Site Conditions and Precautions

Ensure that systems remain fully connected and operational at all times, except where shutdowns are required.

Where shutdowns of systems are required:

- Provide the Superintendent and Owner with written notification at least 5 days in advance of the intended shutdown time.
- Keep each shutdown time to a minimum.
- For each shutdown, complete the works and ensure the systems are fully operational prior to leaving the site for the day.

Carry out the work in such a manner as to interfere as little as possible with other trades and persons having access rights.

Ensure that normal access ways within the buildings are unobstructed.

Ensure to check existing site services, both underground and overhead, prior to commencing work, and verify existing and proposed services are located in accordance with drawings prior to commencement. Plan work practices based on these existing conditions to minimise risk and maximise workplace safety.

Wherever possible locate all existing services prior to excavation or working on existing structures.

Allow to protect installed services during the construction phase, services installed in-ground shall wherever possible be installed at a depth that will provide sufficient depth of cover and protection from heavy construction equipment and vehicles. Allow to coordinate these requirements with the builder.
(eg the builder may need to provide additional temporary ground cover over services and/ or signage to protect services where additional cover is not possible)

Specified grates & frames may not suit construction loads. Allow to protect these items. All grates & frames damaged during the contraction stage shall be replaced without incurring additional costs to the contract.

3.19 Access to Site

Generally, access to the site, working hours and storage facilities shall be agreed with the Superintendent at construction commencement. However, certain restrictions may be applicable to the project and the Sub-contractor shall include within his tender for any restrictions on access.

3.20 Packing Storage and Protection

Plant equipment, apparatus, materials and parts shall be delivered to the site in as- new condition and properly packed and protected against damage due to handling or adverse weather or other circumstances.

Arrange suitable storage areas for materials, plant and equipment. Storage shall be on purpose made racks, mounts, etc., to ensure no damage or deformation of items.

Any items suffering damage in transit or whilst on site shall be rejected and replaced without cost to the Contract.

3.21 Maintenance Access

Locate all relevant equipment to ensure adequate maintenance access in accordance with Workplace Health and Safety regulations.

3.22 Construction Phase Procedures

3.22.1 General

Comply in all respects with procedures established by the Superintendent for the construction phase of the project.

3.22.2 Line of Communication

Ensure compliance with formal transmission of emails, RFIs and submissions through the agreed line of communication.

Advice received by other avenues will not be considered as a formal instruction.

3.22.3 Variations

Variation price submissions shall be accompanied by a detailed description of the work involved. Provide supporting documentation where necessary and as requested.

All costs shall be submitted on a spreadsheet showing itemised break-ups of all materials and labour.

4. DESIGN CRITERIA

4.1 General

This section outlines the general D&C design criteria and requirements and is provided to assist the tender. Other sections of the document shall be read in conjunction.

The design and submission for approval of the hydraulic services drawings shall be carried out by a fully qualified QBCC licensed Hydraulic Consultant. The installation shall be installed by a fully qualified QBCC licensed plumbing contractor.

4.2 Cold Water Service

4.2.1 Infrastructure

There is an existing authority water main located within street frontage. Allow to connect in accordance with the local water authorities' requirements. Refer to civil drawings for master meter requirements.

4.2.2 Proposed Design

- Provide a new metered potable water service throughout the project as required and in accordance with the minimum design requirements and as indicated within the hydraulic intent drawings.
- The cold water system shall comprise the design, supply and installation of cold water pipework and associated valves reticulated to all relevant sanitary fixtures, fittings and tapware in accordance with AS 3500.1.
- Pipe sizing shall be noted to maintain velocities to a maximum of 2.4 m/s and shall be sized to suit the projects operational flow rate requirements.
- Where not shown on the drawings, all cold water service branches shall be a minimum of 15mm diameter with a maximum length of 6 metres for one fixture and 20mm diameter for two or more.
- Minimum supply pressures of 250 kPa shall be provided to all fixture outlets, with pressure reduction being incorporated as required to ensure pressures are limited to 500 kPa to any outlet.
- Within the building all pipe work shall generally be installed at high level through the ceiling space.
- The cold water system shall be provided with isolation valves for, shut down and maintenance, at the following locations, as a minimum & they shall be readily accessible;
  - In-coming supply to each room containing sanitary fixtures, fittings and tapware or at localised TMV stations.
  - At each fixture or fitting.
- Ensure the potable water supply remains uncontaminated. Review all potential cross-connection risks and provide back-flow devices as required to meet AS 3500 requirements, including Vacuum Breakers to all hose taps.
- Allow to provide a suitable metering system that meets local authority & BAC requirements.
- Allow to provide adequate water supply for use by mechanical services, for wash down complete with backflow

4.3 Hot Water Service

4.3.1 Proposed Design

- Provide a potable hot water service throughout the project as required and in accordance with the minimum design requirements and as indicated within the hydraulic intent drawings.
- The hot water system shall comprise the design, supply and installation of hot water units, pipework, insulation, and associated fittings reticulated to all relevant sanitary fixtures, fittings and tapware in accordance with AS 3500.4.
- The selected hot water units shall be sized to meet the expected maximum hot water demand for each area served.
- Pipe sizing shall be noted to maintain velocities to a maximum of 2.4 m/s and shall be sized to suit the projects operational flow rate requirements.
- The hot water system shall be provided with isolation valves for, shut down and maintenance, at the following locations, as a minimum & they shall be readily accessible;
  - In-coming supply to each room containing sanitary fixtures, fittings and tapware or at localised TMV stations.
  - At each fixture or fitting.
- Thermostatic Mixing Valves (TMV's) shall be provided to all amenities throughout the project, ensure TMV's are set at 45°C & dead legs shall limited to 10m.
- Minimum supply pressures of 250 kPa shall be provided to all fixture outlets, with pressure reduction being incorporated as required to ensure pressures are limited to 500 kPa to any outlet.
- Ensure the potable water supply remains uncontaminated. Review all potential cross-connection risks and provide back-flow devices as required to meet AS 3500 requirements.

4.4 Fire Hydrant & FHR Service

4.4.1 Infrastructure

There is an existing authority water main located within street frontage. Allow to connect in accordance with the local water authorities' requirements. Refer to civil drawings for master meter requirements.

4.4.2 Proposed Design

- Provide a new Fire Hydrant & Fire Hose Reel service as required to meet base build fire hydrant & fire hose reel coverage requirements as required and in accordance with the minimum design requirements and as indicated within the hydraulic intent drawings.
- Ensure sufficient and compliant valving, signage and booster assembly is provided to meet AS 2419, QFD & ARFF requirements.
- Pipe sizing shall maintain velocities below 4 m/s, shall ensure maximum pressure loss of 150 kPa during QFD boosting is not exceed and pipe sizing shall be sized to suit the projects combined operational flow rate requirements.
- Provide capped stubs for possible tenancy fit out, where required.
- Coordinate lockable (003) keyed Access Gate points around the fence line perimeter to ensure the QFD have ready access to the building.
- Allow for installation (VAD) Visual Alarm Device to AS 2419.1 & Fire Engineering requirements

4.5 Sanitary Plumbing and Drainage

4.5.1 Infrastructure

There is an existing authority sewer main located within street frontage, Allow to connect in accordance with the local authorities' requirements. Refer to civil drawings for proposed sanitary drainage connection point.

4.5.2 Proposed Design

- Provide a sanitary plumbing and drainage system as required and in accordance with the minimum design requirements and as indicated within the hydraulic intent drawings.
- The sanitary drainage and plumbing system shall comprise the design, supply and installation of pipework, vents and associated fittings reticulated to all relevant sanitary fixtures in accordance with AS 3500.2. Ensure clear-outs to surface are provide throughout the design.
- Allow for the installation of suitable swivel and expansion joints complete with detailing to meet the Ys valves for the local soil conditions and in accordance with AS 2870 & local authority requirements.
- Ensure sufficient under slab fixings are allowed for
- Overflow relief gullies shall be provided to minimise surcharge into the buildings. The installation shall be in accordance with the requirements of AS 3500.2.

4.6 Trade Waste Plumbing and Drainage

4.6.1 Proposed Design

- Provide a trade waste plumbing and drainage system as required and in accordance with the minimum design requirements and as indicated within the hydraulic intent drawings.
- The TW drainage and plumbing system shall comprise the design, supply and installation of pipework, vents and associated fittings reticulated to all relevant fixtures in accordance with AS 3500.2 & the trade waste authorities requirements. Ensure clear-outs to surface are provide throughout the design.
- Allow for the installation of suitable swivel and expansion joints complete with detailing to meet the Ys valves for the local soil conditions and in accordance with AS 2870 & local authority requirements.
- Allow to size and install approved Grease Arrestors to match the number and type of selected fixtures in accordance with AS 3500.2, manufactures and the local authority requirements. A minimum GT size has been noted @ 5000L for each Tenancy
- Ensure sufficient under slab fixings are allowed for
- Ensure ease of future maintenance is considered in regard to the locations of GT's

4.7 Stormwater & Rainwater Drainage

- Infrastructure

Refer to civil engineers' drawings for details.

- Proposed Design

- Provide and extend hydraulic stormwater discharge via in-ground pipework to the civil stormwater system as required. Refer to civil drawings for details.
- Allow for the installation of suitable swivel and expansion joints complete with detailing to meet the Ys valves for the local soil conditions and in accordance with AS 2870 & local authority requirements.
- Provide and size guttering and downpipes to AS 3500.3 requirements as required. Eaves guttering shall sized to 1: 20yr ARI events to suit the local requirements.
- Box Gutters shall be sized to 1: 100yr ARI events to suit the local requirements & AS 3500
- Provide Hail Guards to all guttering, to BAC requirements.
- Ensure sufficient under slab fixings are allowed for.
- Ensure Slip Resistant, Heel Guard and Load Rated outlets & grates are to provided where required & indicated in the design intent drawings.

5. MATERIALS

5.1 General

All materials shall be new, first quality and the best of their respective kinds. Second quality or inferior materials shall be rejected. All costs associated with replacement of rejected materials shall be borne by the Sub-contractor. All materials shall conform to the latest Australian Standard Specification, Code or Interim Code. If no Australian Standard exists they shall conform to the latest British Standard or the American Society for Testing and Materials in that order.

If materials or products are supplied by the manufacturer in closed or sealed containers or packages, bring the materials or products to point of use in the original containers or packages.

For the whole quantity of each material or product use the same manufacturer or source and provide consistent type, size, quality and appearance.

Identification of a proprietary item within the specification does not necessarily imply exclusive preference for the item so identified, but indicates the minimum necessary properties of the item. If alternatives are proposed, submit proposed alternatives and include samples, available technical information, reasons for proposed substitutions and cost savings. State if use of proposed alternatives will necessitate alteration to other parts of the works.

Transport, deliver, store, handle, protect, finish, adjust, prepare for use, and use manufactured items in accordance with the current written recommendations and instructions of the manufacturer or supplier.

Orders for materials shall be made as soon as practicable to ensure delivery of materials are received to meet the construction program.

Submit the recommendations and instructions, and advise of conflicts with other requirements.

Advise of activities that supplement, or are contrary to, manufacturers or suppliers' written recommendations and instructions.

If products must comply with product certification schemes, use them in accordance with the certification requirements.

All materials and equipment shall have a verified history of successful use in the commercial environment. No prototype materials or equipment will be accepted

All items in the section shall be approved by the local authority and bear WaterMark.

5.2 Quality of Materials and Workmanship

All materials shall be new, of the best quality and of the class most suitable for the purpose specified.

All equipment shall have a verified history of successful use in the commercial environment. No prototype equipment will be accepted.

All work on the site shall be supervised by a competent licenced tradesman who shall be present on site when requested by the Superintendent, for all phases of the project. A representative shall be nominated, who shall be present and responsible for the day to day running of the project.

All workmanship and materials shall be installed in a neat and tidy manner. Services shall be installed horizontally and vertically plumb and level to the satisfaction of the Hydraulic consultant.

Straight full lengths of tube and pipework shall be used. Short multiple off cuts and lengths of pipework shall not be joined to form a single length.

Where works do not comply with the above workmanship they shall be removed and correctly reinstalled at no cost to the client.

5.3 Material Installation & Training

All Persons carrying out jointing procedures for materials shall be full trained in their specific use and carry appropriate manufacture accreditation where available.

All tools used in the installation & joining process shall be in a fit and working order and comply with the manufacture's recommendations according to operational tolerances and servicing intervals.

5.4 Environmental suitability

It is the sub-contractor's responsibility to ensure all installed materials and components are suitable for the designated location, Greenstar and environmental conditions prior to installation.

Where exposed to weather or external elements all materials shall be selected to be UV resistant and / or protected from the elements by an additional finish (e.g. painted, enclosed, galvanised, sleeved etc) alternatively the sub-contractor shall nominate an alternate suitable material.

5.5 Standards

All items shall be at least equal to or better than the appropriate current Australian Standard. Any proposal to install alternative items to those specified shall be accompanied by a written confirmation by the manufacturer that the proposed item complies in each and every respect to the relevant Australian Standard.

Where some doubts exist as to the appropriate standard a decision shall be made by the Hydraulic Consultant before commencement of any work on or off the site. If any doubt exists as to whether a section of the design is able to comply with the relevant authorities' regulations the Head contractor shall be notified prior to commencement of any work. No consideration of claims for redundant work shall be given if the Head contractor is not notified.

5.6 Fixtures, Fittings and Tapware

Refer to Architects Specification / Schedule of Finishes for all details of sanitary fixtures, fittings and tapware. Allow to take delivery, store, unpack and install all items as listed complete with ancillary fixings, brackets, seals and connections as required to complete the installation in accordance with the manufacturer's requirements.
It is the sub-contractor's responsibility to check all items are Watermarked and meet WELS ratings and Australian standards.

Where required, provide and install adequate concealed supports and noggings to securely fit-off sanitary ware, fixtures, fittings and tapware.

Refer to Architects drawings to exact location of fixtures, fittings and tapware.

5.7 Pipework Materials

5.7.1 Copper Tube & Fittings – Water & Services

- Comply with AS 1432 & AS 3688
- Installed in Type 'B' (unless noted otherwise)
- Installed using straight full lengths of hard drawn lengths of tube.
- Installed in a neat and tidy manner.
- Be approved by the local authority.
- Have WaterMark approval.
- Bare the standards mark for use on water services.
- Fittings shall be de-zincified brass or copper.
- Fitting minimum wall thickness shall not be less than the pipe wall thickness it serves.
- Fittings shall be '<u>Viega Propress'</u> for water and '<u>Viega Propress G'</u> for gas or Approved Equal.
- Joined using the 'Viega propress fit system' in accordance with the manufactures requirements.
- Fittings shall be pre-manufactured and not formed into the pipe.
- Where used in solar applications the appropriate solar rated seal and fittings shall be utilised.

5.7.2 (PE-X) Poly Pipe & Fittings – Water Services (15mm & 20mm internal <u>rough-in only</u>)

- Installed in Cross Linked Polyethylene (PE-X) pipe and fittings - REHAU Rautitan
- Comply with AS 2492 & AS 2537
- Installed in a neat and tidy manner.
- Be approved by the local authority.
- Have WaterMark approval.
- Bare the standards mark for use on water services.
- Be of the same manufacture for pipe and fittings
- Fittings shall be de-zincified brass or copper.
- Fittings REHAU Rautitan Approved Equal.
- Fittings shall be pre-manufactured and not formed into the pipe.
- Shall <u>NOT</u> be used in the following areas: (Note: Cu type B shall be used instead of in these areas)
  - - solar applications

- - within 1m of a hot water unit
- - located externally or in contact with UV radiation (sunlight)
- - water meter manifolds
- - where visually exposed

5.7.3 PE (polyethylene) Pressure pipe PE100 SDR 11 – External Water Main Services (in-ground)

- Be installed using 'Vinidex BLUE Stripe'
- Comply with AS 4130 & AS 3500
- Be installed in Class PN 16 (minimum)
- Be approved by the local authority
- Have WaterMark approval.
- Shall not be installed within or under buildings or above ground
- Fittings shall be pre-manufactured and not formed into the pipe.
- Incorporate fittings of the same manufacture to the pipe used.
- Electrofusion weld jointed in accordance with manufacture instructions.
- Installed & bedded, with thrust blocks as required to suit manufacturers requirements.
- NOTE: Pipe sizing shall meet the design documents requirements, all size sizes shown on drawings are based on nominal diameter (DN) copper type B sizing. Where installing PE pipework in-ground, sizes larger then shown on the drawings shall be installed. For example:

  - **20mm shown on drawings = 25mm OD PE 100 SDR 11 (PN 16)**
  - **25mm shown on drawings = 32mm OD PE 100 SDR 11 (PN 16)**
  - **32mm shown on drawings = 40mm OD PE 100 SDR 11 (PN 16)**
  - **40mm shown on drawings = 50mm OD PE 100 SDR 11 (PN 16)**
  - **50mm shown on drawings = 63mm OD PE 100 SDR 11 (PN 16)**
  - **65mm shown on drawings = 75mm OD PE 100 SDR 11 (PN 16)**
  - **80mm shown on drawings = 90mm OD PE 100 SDR 11 (PN 16)**
  - **100mm shown on drawings = 125mm OD PE 100 SDR 11 (PN 16)**
  - **150mm shown on drawings = 180mm OD PE 100 SDR 11 (PN 16)**

5.7.4 PE (polyethylene) Pressure pipe PE100 SDR 11 – External Fire Main Services (in-ground)

- Be installed using 'Vinidex RED Stripe'
- Comply with AS 4130 & AS 2419.1
- Be installed in Class PN 16 (minimum)
- Be approved by the local authority
- Shall not be installed within or under buildings or above ground
- Fittings shall be pre-manufactured and not formed into the pipe.
- Incorporate fittings of the same manufacture to the pipe used.
- Electrofusion weld jointed in accordance with manufacture instructions.
- Installed & bedded, with thrust blocks as required to suit manufacturers requirements.
- NOTE: Pipe sizing shall meet the design documents requirements, all size sizes shown on drawings are based on nominal diameter (DN) copper type B sizing. Where installing PE pipework in-ground, sizes larger then shown on the drawings shall be installed. For example:

  - **100mm shown on drawings = 125mm OD PE 100 SDR 11**
  - **150mm shown on drawings = 185mm OD PE 100 SDR 11**
  - **200mm shown on drawings = 250mm OD PE 100 SDR 11**

5.7.5 Steel Pipe & Fittings – Fire Hydrant, Hose Reel & Test Drain Services (above ground)

- Comply with AS 1074 & AS 2419
- Be installed in 'Medium weight' galvanised steel pipe and fittings.
- Installed in a neat and tidy manner.
- Be approved by the local authority.
- Bare the standards mark for use on fire services.
- Fittings shall be joined using 'Victaulic' couplings via rolled grooving methods.
- Fittings and pipework wall thickness shall be suitable to exceed the expected testing and operating pressures for the fire service.
- Fittings shall be pre-manufactured and not formed into the pipe.
- The sub-contractor shall verify correct roll grove gauge depths prior to commencement of works, in accordance with 'Victaulic' manufacture requirements.

- The use of Gal in-ground shall be kept to a minimum and shall not exceed 1.5m in length and shall be double-wrapped in petrolatum tape.
- Where Flanges are used ensure minimum Table E or higher is utilised, as required to suit intended test and operational pressure requirements

5.7.6 uPVC (DWV) Pipe & Fittings – Sanitary Plumbing & Drainage Services

- Comply with Best Environmental Practice (BEP PVC) requirements in accordance Greenstar and ESD project requirements
- AS 1260 & AS 2032
- Be approved by the local authority.
- Bare the standards mark for use on drainage systems.
- Fittings shall be pre-manufactured and not formed into the pipe.
- Laid in trenches without sags or deflection greater then 5mm
- Solvent weld jointed in accordance with manufacture instructions.

Minimum SN rating and depth of maximum depth of ground cover over pipework			
100mm DWV - SN6 Max cover 4 meters	150mm DWV - SN4 Max cover 4 meters	225mm DWV - SN4 Max cover 4 meters	300mm DWV - SN4 Max cover 4 meters

Note: Where depths exceed the above, consult with the pipework manufacture and allow for increase of SN rating as required.

A.1.1 HDPE (High Density Polyethylene) Pipe & Fittings – Trade Waste Plumbing & Drainage Services

- Comply with AS 2033 & AS 3500
- Be of equal manufacture to ‘Geberit’ pipes.
- Be approved by the local authority.
- Bare the standards mark for use on drainage systems.
- Fittings shall be pre-manufactured and not formed into the pipe.
- Laid in trenches without sags or deflection greater then 5mm
- Joined using fusion welded fittings in accordance with manufacture instructions.
- Incorporate fittings & expansion joints of the same manufacture to the pipe used.
- HDPE is a reactive material with thermal expansion or contraction caused through temperature differences, with an average linear expansion coefficient of 0.2mm/m°C. The contractor shall allow to install sufficient expansion joints and suitable ‘sliding brackets’ as required. Where in doubt contact the manufacture for installation details.
- Where embedded in concrete, follow manufactures recommend concrete embedded installation requirements.

5.7.7 uPVC (DWV) Pipe & Fittings – Downpipes & Stormwater Drainage Services

NOTE: Refer to Architectural documents for all downpipe materials, allow for the material documented by architecture, over and above uPVC.

Where no materials have been referenced within the architectural documents, allow for uPVC as note below.

- Comply with Best Environmental Practice (BEP PVC) requirements in accordance Greenstar and ESD project requirements
- AS 1260 & AS 2032
- Be approved by the local authority
- Bare the standards mark for use on drainage systems.
- Fittings shall be pre-manufactured and not formed into the pipe.
- Laid in trenches without sags or deflection greater then 5mm
- Solvent weld jointed in accordance with manufacture instructions.
- NOTE: Refer to BAC requirements & details for all downpipes installed up to 2000mm above finished floor level. These low-level sections for downpipes shall be protected from future vandalism and shall be provided with bollard & sheaths up to 2000mm high to all downpipes
- Shall discharge over Sumps & Grates where noted complete with 75mm clearance between grate and base of downpipe.

Minimum SN rating and depth of maximum depth of ground cover over pipework			
100mm DWV - SN6 Max cover 4 meters	150mm DWV - SN4 Max cover 4 meters	225mm DWV - SN4 Max cover 4 meters	300mm DWV - SN4 Max cover 4 meters

Note: Where depths exceed the above, consult with the pipework manufacture and allow for increase of SN rating as required.

5.7.8 RCP (Reinforced Concrete Pipe) Pipe & Fittings – Stormwater Drainage Services

- Utilised for pipework 375mm and larger
- Shall be equal to 'VSuperTite' by VantagePipes.
- Comply with AS 1646, AS 3725 & AS 4139
- Be approved by the local authority.
- Bare the standards mark for use on drainage systems.
- Fittings & Couplings shall be pre-manufactured and not formed into the pipe.
- Be of the same manufacture for pipe, fittings & couplings
- Laid in trenches without sags or deflection greater then 5mm
- Be of minimum class 2 pipe.
- Joined using rubber ring joints & lubricant in accordance with the pipe manufacturer's requirements.

5.7.9 Cast-iron Pipe & Fittings –Stormwater Drainage Services (where minimum cover 100mm to 300mm required)

- Be installed using "Dimax Tyton Xtreme Z+" by Viadux
- AS 2280.
- Be approved by the local authority.
- Bare the standards mark for use on drainage systems.
- Fittings shall be pre-manufactured and not formed into the pipe.
- Laid in trenches without sags or deflection greater then 5mm
- Joined using rubber ring joints & lubricant in accordance with the pipe manufacturer's requirements.
- All pipework and fittings shall be wrapped in a polyethylene continuous sleeve, refer to 'Installation - Protection of In Ground Metal Piping' and manufacturer's requirements.

5.7.10 PVC-U Pressure Pipe & fittings – Rising Main & Pump Out Risers

- Comply with AS 1477, AS 4020 & AS 3500
- Be installed in Class PN 12 or PN 16 (minimum)
- Be approved by the local authority
- Fittings shall be pre-manufactured and not formed into the pipe.
- Incorporate fittings of the same manufacture to the pipe used.
- Solvent weld jointed in accordance with manufacture instructions.

5.7.11 PE (polyethylene) Pressure pipe PE100 SDR 11 – Rising Main Sewer Services (in-ground)

- Be installed using 'Vinidex CREAM Stripe'
- Comply with AS 4130 & AS 3500
- Be installed in Class PN 16 (minimum)
- Be approved by the local authority
- Shall not be installed within or under buildings or above ground
- Fittings shall be pre-manufactured and not formed into the pipe.
- Incorporate fittings of the same manufacture to the pipe used.
- Electrofusion weld jointed in accordance with manufacture instructions.
- Installed & bedded to suit manufacturers requirements.
- NOTE: Pipe sizing shall meet the design documents requirements, all size sizes shown on drawings are based on nominal diameter (DN) copper type B sizing. Where installing PE pipework in-ground, one size larger then shown on the drawings shall be installed. For example:
    - 20mm shown on drawings = 25mm OD PE 100 SDR 11, etc..

5.7.12 Chrome Pipework - Fixture Connections

- Shall Be used for all pipework fixture connections exposed to view within the building.
- Incorporate a cover plate between the pipe waste or water connection.
- Joined with screwed rubber compression fittings for waste pipes.
- Not be joined into the drainage pipes with silicone sealants.
- Free of defects and be of first quality chrome plating.
- Bare pipework requiring chrome plating shall be manufactured free of dents, creases or any material that will affect the chrome plating process.

5.8 Valves

Unless otherwise indicated all valves shall be:

- In accordance with the respective Australian Standard and carry either Watermark or AGA Certification where used for gas services.
- Of the same model and manufacturer.
- Of bronze / copper alloy (DR) dezincification resistant material up to and including 80mm in diameter.
  Note: All valves and fittings were installed into a stainless-steel pipework system shall also be stainless steel 316
- Be installed in an accessible position for means of operation and/or removal.
- Up to and including 50mm diameter - be of a screwed pattern with a barrel union fitted to the outlet side of each valve.
- 65mm diameter and above - be of a flanged table E type fitted to the outlet side of each valve 65mm and above.
- Valve sets 50mm and larger installed above ground shall have their weight supported via either appropriate suspended clipping or through the use of a ground supported galvanised steel frame.
- All valves shall be suitable for temperatures up to 90°C.

Location of valves shall:

- Be positioned to ensure ease of access and maintenance.
- Be coordinated on-site with other services to ensure they are not located in inaccessible locations.
- Where located in-ground, be provided with a suitable load rated path box with riser to FSL complete with valve identification incorporated into the lid.

5.8.1 Control Valves - Water

15mm to 20mm: Jumper Valve

- Loose Jumper Valve type - fitted with "O" ring seals to the spindle OR
- Manufactured and tested in conformity with AS 1718 - Water Supply - Copper Alloy Screw-Down Pattern Taps - Specified by Dimensions.
- Constructed of brass materials.
- Fitted with unions on each side of the valve for maintenance/removal. Alternatively one union can be deleted provided an approved screwed joint is substituted.
- Recessed stop/control valves shall be chromium plated all brass construction, of pattern and finish equal to that specified under Taps and Valves.

32mm to 50mm: Ball Valve

- Ball Valve type - Straight through flow, solid wedge type.
- Full bore pattern with handle parallel to direction of flow when fully open.
- Bronze bodied with hard chromed brass or stainless steel ball, stainless steel stem and PTFE seat.
- Screwed connections.

65mm to 150mm: Gate Valve - (Table 'E' Flanged)

- Gate Valve type - Straight through flow, solid wedge type.
- Bronze bodied to AS 1628–1999 or cast iron to AS 3579–1993 or cast steel with zinc free, non-rising inside screw type spindles.
- Medium pattern.
- Replaceable bronze seat.
- Flanged connections.

5.8.2 Control Valves – Fire Hydrant Services

Above Ground - 100mm to 150mm (flanged or rolled grove)

- Full-flow outside screw and yoke wheel gate valve - indicating type, to AS 2419.
  or
- Low torque wheel-operated multi-turn post indicator butterfly valve with all metal actuating mechanism to AS 2419.
- All shall be secured and locked in the open position (via steel chain, with 003 keyed lock)
- Have affixed to the valve a plate inscribed with "FIRE MAIN VALVE – SECURE OPEN" in letters not less than 8mm high & in accordance with AS 2419 & local fire brigade requirements.
- Where fire service isolation valves are installed with 'pre-installed monitoring wiring' & are not required to be wired to the FIP. Allow to remove all excessive unwanted wiring and provide a suitable weatherproof cap.
- All above ground ring-main isolation valves, shall be installed **no higher than 2m** above FFL to satisfy the intent of "ready access"

Note:

¼ turn Butterfly valves <u>are not</u> an acceptable valve on fire services, nor are they acceptable as part of the backflow valve assembly, prior to the fire service.

5.8.3 Check Valves

- Approved by Local Authority.
- Manufactured and tested in conformity with AS 1628
- Check valves on water services 20mm diameter and under shall be spring-loaded check valve manufactured with corrosion resistant components.
- All other check valves shall be of a horizontal pattern fitted with a limit stop to prevent sticking in the open position.

5.8.4 Backflow Prevention Valves

- Comply with AS 2845 & AS 3500
- Where required ensure backflow valves discharge over a suitable drainage outlet, complete with air gap.
- Allow to maintain all registered testable backflow devices for the first 12 months & include the first 12 month maintenance test to all registered valves.
- Where RPZD's are installed, they shall be installed above a tundish, complete with air gap (2 x times the OD of the RPZD)

5.8.5 Reflux Valves

- Shall be equal to "Plastec"
- Shall be inclusive of a riser and bolted trap screw lid finished at FSL, to allow for easy access and servicing.
- Allow to identify and provide labelling to the bolted trap screw lid.

5.8.6 Pressure Reduction Valves

- Confirm to AS 1357
- Shall be fully adjustable, complete with in-built strainer.
- Shall maintain constant pressure reduction, with no pressure creep.
- Shall have a maximum reduction ratio of 3:1, where greater reduction is required, allow for staged reduction via multiple valves.
- Shall be sized to suit the minimum and maximum expected flow rates. Allow to install a suitably sized smaller PRV by-pass arrangement where low flows could be expected, to assist in noise reduction.
- Where installed as part of high-rise pressure reduction station, allow to install dual pressure reduction valves, complete with IV's pressure gauges and unions either side of the PRVs to allow for removal and future servicing.

5.8.7 Strainers

- Confirm to AS 1357
- Strainers shall be 'Y' type and fitted upstream of all back-flow prevention devices, TMV's and solenoid valves.
- They shall contain a removable screw cap for strainer access and cleaning
- Mesh size shall be a minimum of 500 micron

5.8.8 Thermostatic Mixing Valve (TMV)

- Comply with AS 4032 & AS 3500
- The operating outlet temperature, shall be set at **43°C**
- The valve to be used must be approved for use in Health Care Facilities.
- The valve must have a wax element which gives complete fail safe and have built in stop cocks for service ability.
- The valve shall be, equal to 'Enware – Aquablend 1500' and contain built-in 'thermal flush capability' to allow hot water disinfection through the outlets and valve.
- The thermostatic mixing valves will not be placed any further than **6** lineal metres of pipe from the outlet of the fixture it is servicing and shall not be placed in such a position which would make it subject to direct sun or in such a position where it could be subject to heat conversion from any other apparatus.
- All valves shall have their temperature set and tested following completion of works and all test results shall be recorded and included within the Instruction and Maintenance Manual. Testing shall include failsafe testing with isolation of cold water service.
- Allow to provide in-wall matching manufacture Stainless Steel lockable cabinets and signage.
- All valves shall be tagged and numbered. Provide a valve schedule of the installation on completion.

5.8.9 Recycled Valves

- Recycled or reused valves of any type will not be accepted.

5.9 Flanges

Unless otherwise indicated all flanges shall:

- Confirm to AS 2129, AS 2528 & AS 4087
- Suit the maximum testing pressure of the associated service and be a minimum of 'table E'
- Brass flanges for copper tubing
- Galvanised mild steel flanges for galvanised mild steel
- Cast-iron flanges for cast iron pipes
- PVC flanges for PVC piping

5.10 Flexible Connectors

Water services
Where used, flexible connectors shall be equal to 'Abey – Polyamide Hi Class Hooker' and shall comply with AS 3499 and be Watermark approved.

Ensure the appropriate lengths of flexible connectors are used, where excessive length hoses have been utilised, the hydraulic sub-contractor shall replace them at no additional cost to the client and as directed.

The external braiding shall be polyamide and incorporate an anti-kinking design.

Ensure to include within the O & M manual the manufactures recommend service and replacement intervals.

5.11 (PWD) Shower Hose Restrictor Device

Provide and install a suitable shower hose restrictor device or bracket, in accordance with AS 1428.1 requirements. The restrictor shall prevent the shower head from reaching into the toilet pan, basin or onto the floor.

Ensure to coordinate with builder for sufficient noggings and or timber framing behind walls, to ensure a solid and secure fixing point for restrictor device.

Shower hose restrictor brackets shall be equal to "Con-serv" Model: HR 015 C – chrome.

5.12 Path Boxes

Supply and install Valve / Path boxes to provide access to all underground valves.

The boxes shall be cast iron boxes with hinged covers (ensure the valve box selected is suitable to suit maximum expected vehicular load.) Set tops flush with pavement surfaces, or 10 mm above unpaved surfaces. Provide an in situ concrete surround 150 mm thick, to the top, bottom and sides of the box, with exposed surface steel trowelled. Set a shaft of pipe beneath each box to provide access to the valve & handle to ensure free operation. Stamp the box cover with the word applicable to service concerned with the isolation valve below.

Where local authorities have requirements for water meter path boxes, allow to ensure local authority requirements are meet.

5.13 Silicone Sealant

Silicone sealant shall be self-polishing with anti-fungicide additive and used as recommended by the manufacturer. White shall be used around vitreous china sanitary ware and clear for seal under fixture taps and stainless steel, Unless told otherwise

5.14 Adhesives

Provide adhesives and sealants capable of transmitting imposed loads, sufficient to ensure the rigidity of the assembly or the integrity of the joint or label and which will not cause discolouration or lack of adhesion of finished surfaces.

5.15 Concrete

Concrete shall be of minimum 25 MPa strength, or as directed by the structural engineer. All site mixed concrete shall be of 4:2:1 mix. Mortar shall be 2:1 cement mix. Waterproof render shall be 3:1 mortar waterproofed, with approved brand of waterproofing compound used directly in accordance with the manufacturer's direction

5.16 Acoustic Lagging

Provide and install acoustic lagging to all suspended sanitary & stormwater drainage pipework, in accordance with applicable NCC requirements and all acoustic lagging shall comply with the current NCC acoustic requirements, with specific reference being made to NCC volume 1, part F5.

Where an acoustic engineering report has been undertaken and is applied to the project, the highest acoustic requirements found between either the NCC or within the acoustic engineering report shall be followed and will overwrite the below minimum requirements.

Apply acoustic lagging after testing & inspection of pipework as has been completed.

The acoustic lagging for the soil, waste and stormwater pipes shall be equal to Thermotec “NuWrap 5”. The Barrier shall be a barium loaded limp mass polymer with a density of 5kg/m2. The decoupling layer shall be an open cell Polyether-urethane convo-luted foam of 25mm maximum thickness with a reinforced aluminium foil facing. The lagging must demonstrate to meet GreenStar, Green Building Council and Dubai Municipality low VOC requirements and be able to operate continuously at a maximum temperature of 100 degrees Celsius.

When installing, material should be cut to size and fixed in place, with no gaps, using a self-adhesive aluminium foil faced tape of 72mm width. Straight lengths of pipe must incorporate a minimum 50mm overlap and again be sealed with a 72mm wide foil tape. All material surfaces to be cleaned prior to affixing tape.

Ensure all manufacturers recommendations and installation requirements are followed and meet.

5.17 Thermal Insulation

Insulation shall not be installed on any pipework until all relevant tests and inspections have been carried out.

Before application, remove all scale, rust, grease and debris from pipework.
All insulation shall be fitted tightly to the surface to which it is applied, without gaps or edges being exposed. End of sections of insulation shall butt to one another over the whole surface to be insulated.

All insulation shall meet and exceed the requirements of the BCA, AS 3500 & AS 1530.3 for fire retardant & thermal properties

5.17.1 Hot Water Insulation - Non-Circulating systems

Internally
Shall be equal to ‘**Thermotec – E-flex ST’**, with preformed sectionalised fire retardant, closed cell polyethylene foam insulation. The following Minimum R valves shall be meet and take into account the ‘climate region’ requirements of AS 3500.4 – Table 8.2.2

- 13 mm thickness R = 0.3 - the majority of QLD areas fall under this value. Contractor to confirm their location and refence AS 3500.4 prior to installation.
- 25 mm thickness R = 0.6
- 38 mm thickness R = 1.0

Externally & where sunlight (UV) may affect pipework & insulation
Shall be equal to ‘**Thermotec – E-flex HT Solar’**, with preformed sectionalised EPDM closed cell foam insulation, using the same R values and minimum thicknesses as noted above.

Note: Where pipework is installed within conduit in-slab. Ensure to install sufficient sized conduit to enable thermal installation to be installed to meet AS 3500.4 Table 8.2.2

5.17.2 Hot Water Insulation - Circulation Systems

Shall be equal to ‘Thermotec – 4-Zero’, with preformed sectionalised fire retardant, closed cell polyethylene form insulation and incorporate a factory applied aluminium foil.

All joints shall be taped using a pressure sensitive aluminium tape, as approved by the manufacture.

Timber insulation blocks / ferrules or approved equivalent shall be utilised to support all pipework at support brackets and shall be equal to size in the adjacent insulation.

The following below R valves & thickness shall be meet and take into account the ‘climate region’ requirements of AS 3500.4 – Table 8.2.2

- **25mm wall thickness shall be applied to all hotwater pipework up to and including 50mm.**
- **30mm wall thickness shall be applied to all hotwater pipework 65mm and above.**

- **Note:** All branches thermal insulation from the flow and return system shall be continued in the same thermal thickness as noted above for a minimum of 500mm, in accordance with AS 3500.4 – Table 8.2.2.

Hydraulic heat loss calculations have been performed using technical data as provided by Thermotec, deviations in terms of thermal performance by not using ‘Thermotec 4-zero’ shall be the responsibility of the contactor.

Where hot water services are encased within stud walls the maximum possible thickness insulation shall be installed complete with glued joints to the manufacture’s recommendations.

NOTE:
All insulated pipe located in plant rooms, located externally or exposed in process areas shall be clad in tight fitting Colourbond metal sheathing with all joints riveted and silicone sealed.

5.17.3 Condensate Insulation

Shall be equal to ‘Thermotec E-Flex ST’, with preformed sectionalised fire retardant, closed cell form insulation.

All condensate drains shall be wrapped with 13mm wall thick insulation and installed in accordance with manufacture requirements.

As a minimum all condensate drainage pipework leading to the trap, (including the trap) and horizontal pipework shall be thermally insulated.

5.18 Dissimilar Materials

All materials including bolts, which may trigger a corrosive or galvanic reaction when in contact, shall be isolated from each other by approved non-metal jointing or insulation.

5.19 Warning Tape & Tracing Wire

All buried hydraulic services shall be identified with a 100mm wide polypropylene plastic tape complete with 0.7 mm AISI 316 grade stainless steel tracer wire and laminated with a printed warning tape on the top surface.

The warning tape shall be placed 150 mm above the top of the service. Ensure where warning tapes need joining that all joints are completed using matching crimps and crimping tool.

Trace wire warning tape should be installed so the start, end and various access points along the length are accessible for connection purposes. Valve boxes, pits and manholes shall be utilised for this purpose. Refer to manufacture for typical installation details and requirements.

The warning tape shall be a continuous length coil and colour coded to AS 1345, in addition to naming the service. e.g. BLUE - water supply below.

Approved warning & tracing tape includes ‘Wavelay detectable underground warning tape’ and is available through suppliers such as “Tapex” or approved equal.

5.20 Identification labels

Identification labels shall be secured to all pipework throughout the entire project, except for vent pipes above roof level & external downpipe pipework.

Before application, remove all scale, rust, dust, grease, and debris from pipework / insulation and ensure identification labels adhere to pipework correctly. The hydraulic sub-contractor shall be responsible to re-apply all loose or missing labels.

Labels shall be installed in accordance with manufactures requirements and in accordance with AS 1345.
Where externally mounted labels shall be weather/UV and corrosion resistant.

Labels are to be of the vinyl, pressure sensitive, self-adhesive type consisting of combined flow direction arrow and name of services and shall be equal to the “Safetyman” pattern.

Labels shall be positioned on both sides of all pipework, valves, bends and junctions and along the entire length of the pipeline at a maximum of 3000 mm centres.

Where pipes are installed within ducts or false ceiling areas, additional labels shall be installed to be visible at the access panel position.

Gas services pipework identification shall include type of gas & operating pressure of the system.

5.21 Valve Identification

All control valves shall be identified by a 1.0mm thick melamine plate minimum dimension of 65mm x 40mm, with the purpose stamped into the plate in minimum 5mm high letters.

Identification plates shall be securely & permanently fixed to the valve.
Example: Male Toilet – Cold Water Isolation

5.22 Clear Out to Surface (COS)

Where indicated on plan (COS) install clear out inspection fittings to finished surface levels.

Risers from pipelines serving COS points shall:

- Extend vertically and be fitted with a suitable in-ground expansion joint to suit the Ys value of the ground movement.
- Be provide a minimum 100mm thick concrete (25mpa) support pad located below each riser
- Terminate with a Brass Or Stainless Steel screw-out COS and frame suited for anticipated loads.
- Be fitted with correct COS point to suit the finished surface material of the given area. (eg Vinyl, Carpet, Tiles, Exposed concrete etc.) Refer to architectural drawings for details of finished surfaces.
- Ensure where COS locations noted within external vehicle areas, within warehouses or areas subject to vehicle, forklift or heavy loads, ensure to provide separate load rated cover and frame, set into a concrete surround suited for the anticipated loads, these items are to prevent loads being imparted onto the COS point

All COS points are to be Stainless Steel or Chrome plated brass finish as per ‘SPS’ (Specialty Plumbing Supplies.)

<u>NOTE</u>: Ensure to review Architectural Specification to ensure architectural project specific COS points are not specified or requested elsewhere in the project schedules.

5.23 Stormwater Trench Grates and Sumps

All trench grates, roof sumps and outlets, sump grates and frames used, shall be manufactured in accordance with AS3996 and be of the required strength to carry the vehicular or pedestrian traffic and shall be selected suit to the intended design flow rates.

All grates used shall be longitudinal bar gratings set flush with surrounding finished levels and shall be ‘Heel Guard’ & ‘Slip Resistant’ type were indicated or located in pedestrian areas.

Ensure that all the pits, trenches, channels or sumps match the load rating of the specified grating. (i.e. Class D Grate, shall also have a matching class D frame and sump.)

Ensure that the ‘slip rating’ of the installed grating matches that of the surrounding pavement / finished surface area

Grates shall also be Finger and Toe resistant and be provided with lockdown capability, set flush to the surrounding area and comply with DETE standards.

Allow to protect installed grates and frames during the construction phase, as grates & frames may not suit construction loads. All grates damaged shall be replaced without incurring additional costs to the contract.

5.24 Rainwater Outlets

Supply and install flat type drainage outlets cast in the concrete structure in positions indicated on the drawings. All grates shall be nickel bronze or stainless steel. Set grates, flush and matching with adjoining surfaces.

All rainwater outlets shall be manufactured in accordance with AS 3996 and be of the required strength to carry the vehicular or pedestrian traffic and shall be selected and installed suit to the intended design flow rates and finished surrounding surface materials.

Ensure that the 'slip rating' of the installed grating matches that of the surrounding pavement / finished surface area

Allow to protect installed grates and frames during the construction phase, as grates & frames may not suit construction loads. All grates damaged shall be replaced without incurring additional costs to the contract.

5.25 Covers and Frames

Covers and frames shall be manufactured in accordance with AS3996 and be 'EJ / Havestock' or approved equal cast iron covers and frames with all edges machine fitted and have removable plastic lifting hole plugs.

Each cover shall have a brass or stainless steel plate bearing the name of the service.

The covers and frames shall be set to the level of the finished surface and filled in with the same material as used for the surrounding path or roadways, lawned areas shall be infilled with concrete. Brass edge trimmed covers and frames shall be used in areas where tile, paver or carpet finish is required.

All covers and frames shall have the manufacturers recommended lubricant applied prior to installation.

All internal covers shall be lock down type and where used to service sanitary or trade waste streams all covers and frames shall be 'air tight'.

Ensure that the 'slip rating' of the installed cover matches that of the surrounding pavement / finished surface area

Allow to protect installed covers and frames during the construction phase, as covers & frames may not suit construction loads. All covers damaged shall be replaced without incurring additional costs to the contract.

5.26 Fixings & Fasteners

Provide masonry anchors, bolts, nuts, washers, screws, threaded rod, nails and plugs as required for installation of the hydraulic services. Materials, finishes and physical sizes shall be as appropriate for the loads, the types of materials in contact with the fastener, fixing and the environment.

Fixings shall comply with AS 5216 and be equal to 'Ramset' 'Dynabolts', 'Dynasets' or equal approved fire rated mechanical expansion type and shall be installed in direct accordance with the manufacturer's instructions, and they shall also be capable of transmitting the loads imposed and be sufficient to ensure the rigidity of the assembly.

Be suitable and selected to ensure full compliance with the requirements of AS 1170.4 for seismic restraint where required.

Where in doubt, contact the manufacture for correct selection of fixings and fasteners to suit the application and installation requirements.

Dissimilar metals: If pipe and support materials are dissimilar, provide industrial grade electrically non-conductive material securely bonded to the pipe to separate them.

- Plastic or nylon type fixings or anchors will not be accepted.
- Shure-drive metal expansion anchors will be accepted on small-bore pipes up to 20mm diameter but shall not be used in suspended or overhead applications.

Stainless Steel
316 grade Stainless Steel components (including fasteners, fixings, anchors and threaded rod) shall be used for all external applications, where:

- Located Externally
- Exposed to weather
- Embedded in masonry
- Located internally and subject to external salt laden (sea breeze) airflow. EG; carparks, open corridors, balconies, wash bays etc
- Where fasting into chemically treated timber

Note: Stainless steel shall not be used in contact with Zincalume or Colourbond steel.

Galvanized steel

Galvanized mild steel components (including fasteners, fixings, anchors and threaded rod) to AS 4680 or AS 1214 as appropriate. And shall be used for all internal applications, installed internally behind the protection of external / masonry walls and roofs.

The exception for external use only being permitted where in contact with Zincalume or Colourbond steel.

- NOTE: Galvanized steel shall not be in contact with chemically treated or 'CCA' / 'ACQ' timber.

5.27 Paint

Paint shall be Dulux manufacture or approved equal and shall be applied in accordance with the manufacturer's instructions. Include for all cleaning, priming or etching as required. Allow for 2 coats of final topcoat paint. All painted finishes shall be streak free and to a professional standard.

5.28 Samples

The following minimum items shall be provided as samples for approval by the Architect prior to commence of works.

- Valve identification plate.
- In-ground Warning Tape.
- Identification Labels.
- Hot Water Insulation.
- Condensate Insulation.
- Rainwater, Trench Grate & Sump Outlets.
- Floor Waste Grate.
- PWD Shower hose restrictor device or bracket.
- Clear Out to Surface (COS) covers for all different floor coverings as required throughout the project, including load rated cover and frames suited for the anticipated loads
- Fixtures and Tapware as requested by the architect.

6. INSTALLATION

6.1 General

All work on the site shall be supervised by a competent fully licenced plumber who shall be present on site for all phases of the project. A representative shall be nominated, who shall be present and responsible for the day to day running of the project.

All workmanship and materials shall be installed in a neat and tidy manner. Services shall be installed horizontally and vertically plumb and level to the satisfaction of the Hydraulic consultant and architect. Provide noggings, brackets, fixings and supports to manufacturer's recommendations and as required.

Where timber is used for supports or noggings, provide cypress pine or similar termite resistant species.

Organise reticulated services neatly and with applicable segregation.

All hydraulic systems and services shall be fully commissioned and verified as working correctly prior to practical handover.

6.2 Safe Working

Prior to undertaking any works, undertake a Risk Assessment / Job Hazard Analysis and conduct a Safe Work Method Statement (SWMS) for all works as required, to meet the specific project requirements and all statuary requirements.

Carry out the work in a careful, secure, safe and tidy manner and take all precautions to ensure the works will not cause nuisance or danger to the public, site personnel and occupants of adjacent buildings.

All work causing inconvenience to others shall have safety lights, barriers, warning signs etc installed and displayed in accordance all standards, codes and regulations.

Unfinished and completed works shall be left in a safe condition during all stages of project.

6.3 Capping Off

During construction, leave all unfinished work in a safe condition and protect the works against damage. At all times open ends of pipes shall be sealed off in such a manner as to prevent the entry of foreign matter into the lines.

Plugs and caps made from rags, wood or paper will not be acceptable for this purpose.

Water services laid in trenches shall be totally sealed to prevent contamination from water within the trench.

6.4 Uniformity

All fittings, accessories and equipment of the same type shall be of the same manufacture and catalogue number.

6.5 Service Connections

As indicated on the attached drawings. Allow all costs to make application, design documentation and construction associated with all services connections.

6.6 Coordination

The Sub-Contractor shall obtain detailed information from other Sub-Contractors, which may affect the works specified and co-ordinate the works as necessary to satisfy the performance and programme.

Provide and be responsible for coordination of all aspects of the work under the hydraulic contract, including but not limited to the preparation and submission of hydraulic shop drawings, installation and commissioning of the hydraulic systems.

The Sub-Contractor shall obtain all dimensions required for locations of tapware and fixtures from Architectural documentation and allow for all required pipe offsets to ensure all services are co-ordinated with other trades, building structure and existing services.

Liaise and co-ordinate with all equipment suppliers to ensure that all equipment is installed and connected in accordance with the installation and functional requirements of the manufacturer. Provide ancillary equipment as required for correct system operation.

6.7 Installation Coordination

Various items of apparatus and equipment will be provided and fixed by others, though-out the build.

The Sub-Contractor shall familiarize themselves with the requirements of the other sub-contractor's works requiring co-ordination and shall examine the plans covering each of these services.

It shall be the responsibility of the Sub-Contractor to schedule work closely so that the work may be installed at the proper time and without delaying the completion of the entire project.

The Sub-Contractor shall carefully check space requirements with other Subcontractors to ensure that the equipment, piping, etc., can be installed in the spaces allotted for same.

When equipment and/or installations associated with the work of the contract are installed by other sub-contractors, the Sub-Contractor shall attend and co-ordinate as necessary.

Associated fittings required to prevent clashes with other trades shall be deemed to have been included in the tender price.

Check all dimensions on site prior to commencing work.

The design drawings do not relieve the Hydraulic Services Sub-Contractor of the co-ordination or buildability responsibility.

All design changes occurring after the letting of the Hydraulic Services Contract, by other trades including architectural, structural and mechanical services, shall be co-ordinated into the installation by the Hydraulic Services Sub-Contractor as necessary.

The Hydraulic Services Sub-Contractor shall be responsible for the installation of all services elements associated with this subcontract, within the ceiling, duct and plants spaces provided in accordance and co-ordinated with all other services trade designs.

6.8 Pipe Locations

Generally, all piping shall be concealed from view where possible within stud walls, cavities or in plumbing ducts unless otherwise indicated. All pipes shall be arranged so as to provide suitable access for maintenance or replacement without disturbing others.

Pipes and fittings shall not be buried in inaccessible locations unless specifically directed by the Superintendent. Allow for all bends and offsets necessary to avoid other services and to meet structural and architectural conditions. Care shall be taken to arrange any exposed pipe work to obtain a neat appearance and true alignment.

Pipes shall be arranged with valves and inspection openings in convenient positions for servicing through access panels provided in walls, ducts, ceilings, etc. Access panels are be provided by the Contractor to suit location of valves. Ensure coordination of final locations are done completed on-site, prior to the installation of all valves and access panels on-site.

Install appropriate unions to valves and fittings to permit easy removal and replacement from pipes.

Pipe work will not be permitted to be chased into masonry walls, unless prior approval granted.

Exposed runs of insulated piping shall be supported so that a clear space of not less than 50 mm is left between the pipe insulation and the nearest wall or ceiling surfaces.

Water service pipes shall not be embedded or cast into concrete structures

6.9 Penetrations & Set Out

Penetrations and Setout of hydraulic works shall be the responsibility of the hydraulic sub-contractor. Coordinate these set out works with other trades.

The hydraulic drawings are produced for general layout and are diagrammatic only. The hydraulic sub-contractor shall make reference to the architectural drawings and fixture / equipment technical data to confirm the exact position for fixtures, plant and services on-site. Rectification of incorrectly located core holes and/or pipework is the hydraulic sub-contractors responsibility.

Each pipe passing through fire / smoke rated walls and floors shall have its own approved penetration.

Pipe sleeves shall be cast into concrete where pipes penetrate concrete columns, beams, floors or walls other than retaining walls.

Coring (drilling) shall be carried out by a firm approved by the Superintendent. The Superintendent shall be notified prior to work being carried out.

Provide flexible material around piping penetrations through building elements. Provide sufficient annular space around the pipe to prevent building elements causing undue stress on any pipe or fitting.

Do not provide additional core holes or fix to the following building elements without prior approval:

(a) Structural building elements including external walls, fire walls, floor slabs and or structural beams

(b) Membrane elements including damp-proof courses, waterproofing membranes and roof coverings

If approval is given to penetrate membranes, provide a waterproof seal between the membrane and the penetrating component. Maintain integrity of building systems such as waterproofing and termite management at building penetrations.

Where services pass through building & tank elements which may allow water ingress, allow to install puddle flanges to all penetrations to prevent water ingress into the building.

Provide written confirmation that all penetrations have been fire/smoke sealed in accordance with the NCC.

6.10 Acoustic Penetrations & Integrity

To any pipe or fitting penetrating an acoustic rated wall or floor ensure the acoustic rating is maintained in accordance with the acoustic engineer’s requirements.

Install hydraulic services to comply with acoustic isolation requirements of the National Construction Code of Australia and statutory requirements.

Note in particular the requirements for residential units and buildings and provide acoustic rated insulation as required to maintain the acoustic rating and in accordance with the acoustic engineer’s report.

6.11 Fire Collars & Smoke Seals

To all pipes and fittings manufactured of a nonferrous material, provide approved fire or smoke seal penetrations using a system to AS 4072.1 & AS 1530.4 and in strict accordance with the manufacturer recommendations and the National Construction Code of Australia to ensure that the required Fire Resistance Level of the building element is maintained.

All retrofit fire collars shall be fixed using steel fixings approved by the testing authority of the fire collar.

All fire collars shall be tested and approved for the proposed application.

On completion and prior to wall or ceiling sheeting being applied all fire collars, smoke & fire penetrations shall be certified by a recognised and registered fire engineer. All fire penetrations shall be identified in the fire penetration schedule. The fire penetration certificate and schedule shall be included within the maintenance manual.

Label all penetrations through fire rated building elements and provide a penetration schedule.

6.12 Piping Installation

Before installation, remove loose scale, burrs, fins and obstructions.

During construction, prevent the entry of foreign matter into the piping system by temporarily sealing the open ends of pipes and valves using purpose-made covers or caps.

- Allow to provide test gates and isolation valves where necessary to suit construction staging and to allow for satisfactory testing of the hydraulic systems.

Install piping in straight lines and at uniform grades with no sags. Arrange to prevent air locks. Provide sufficient unions, flanges and isolating valves to allow removal of piping and fittings for maintenance or replacement of plant. Pipes shall be installed using the longest practical length of tube/pipe to eliminate unnecessary jointing.

Arrange and support piping so that it remains free from vibrations whilst permitting necessary movements. Minimize the number of joints.

Provide at least 25mm clear between pipes and between pipes and building elements, additional to insulation.

Join dissimilar metals using fittings of electrolytically compatible material.

Unless otherwise approved or directed, pipes shall be concealed.

Unless otherwise noted, all piping shall be kept as high as possible to underside of beams, slabs and structural elements so as to maintain maximum head height.

Ensure to install In-ground Warning Tape & Tracing Wire to all services installed in ground, and install Identification labels to all pipework, as previously detailed within "Materials"

Provide access and clearance at fittings which require maintenance or servicing, including control valves and joints intended to permit pipe removal. Arrange piping so that it does not interfere with the removal or servicing of associated equipment or valves or block access or ventilation openings.

Sheath or sleeve metal piping chased into masonry or encased in concrete so that expansion or contraction can take place without damage to the pipe or to the material or surface finish of the surrounding element. Expansion material around pipes shall be not less than 25mm.

Concealed water & Fire service piping shall be maintained under safe operational pressures while subsequent building operations, which may cause damage to the pipes are being carried out.

6.12.1 Fire Hydrant Pipework and Tilt-up Panel Walls

Where Tilt-up Panel Walls are used in the construction of the development, ensure that Fire hydrant pipework is NOT mounted on, pass through, or fixed to tilt-up panel walls.

Where pipework is indicated & positioned close to or adjacent to tilt-up panel walls allow all costs to install suitable free standing pipework support ladder system, in-leu of fixing hydrant pipework to tilt-up panel walls.

6.12.2 Pipework Cast-In Columns, Concrete, Footings & Under Slabs

Where drainage service pipework is to be installed / encased within concrete and structural elements, all drainage pipework shall be HDPE or other approved equivalent with resistance to heat generated from concrete curing and be wrapped in non-absorbent closed cell polyethylene lagging to AS 3500 & AS 2870 requirements.

- Minimum of 10mm thick non-absorbent closed cell polyethylene lagging shall be applied to manufactures requirements to all pipework materials cast in-slab.
- NOTE: All pipework shall be water filled, capped, and tested prior to concrete pour and setting, and shall meet the manufacturer's requirements for cast-in pipework and fittings.
- NOTE: Prior to installation of all cast-in pipework, approval shall be sort via the structural engineer.

Where drainage penetrations of footings are necessary non-absorbent closed-cell polyethylene lagging shall be used around all stormwater and sanitary drainage penetrations through footings. The lagging shall be a minimum of:

- 20 mm thick non-absorbent closed cell polyethylene lagging on Class A to H1 sites
- 40 mm thick non-absorbent closed cell polyethylene lagging on Class H2 and Class E sites.

Also refer to Flexible Joints in Reactive Soil, for additional information

Where water service pipework is to be installed within or under concrete and structural slabs, all pipework shall be installed within suitability sized and watertight conduits and pipework shall be continuous with no joints within or under the slab. The purpose of the conduits is to ensure if the pipework leaks, water will be noticed above the slab and or outside the slab and will not leak unnoticed under the slab.

- NOTE: Prior to installation all of all cast-in pipework, approval shall be sort via the structural engineer.

6.12.3 Pipework within food preparation areas

Where pipework is installed within food preparation areas, ensure AS 4674 installation guidelines are followed.

In particular ensure pipework is either concealed in floors, plinths, walls or ceilings or; fixed on brackets so as to provide at least 25mm clearance between the pipe and adjacent vertical surface and 100mm between the pipe and adjacent surface.
Refer to AS 4674 (Design, construction and fit-out of food premises)

6.12.4 Pipework connected to equipment with heat generation during operation

Where a water service is provided to serve equipment that can generate heat during normal operation, e.g., Ice Machines & Chilled Water Fountains with chiller compressors etc., the final on-site location of connecting pipework should be taken into consideration with regards to the location of heat generating components, to ensure adequate separation is provided to prevent heat transfer to the incoming water supply pipework. (These works assist with control of legionella to such equipment)

6.12.5 Pipework within bushfire areas

Where a water service is installed in a designated bushfire-prone area, all exposed piping shall be metal. Pipes of other materials shall be buried with a minimum depth of cover over the top of the pipe being 300 mm. Ensure to refer to Bushfire Report & AS 3959 for additional requirements

6.12.6 Pipework within contaminated areas

Where a water service is installed in ground that may be contaminated, the service is to be laid in a watertight corrosion-resistant conduit to provide protection for the service.

6.12.7 Pipework within corrosive areas

Where a water service metallic pipe, metallic fittings are installed in a corrosive area, they shall be protected externally, in accordance with AS 3500.1 - 2018 section 5.12.

6.13 Flexible Joints in Reactive Soil

Where Sanitary & Stormwater drainage is being installed within reactive soils, the hydraulic services sub-contractor shall install swivel joints and expansion joints in accordance with the structural engineers, soil survey report, hydraulic details and AS 2870 recommendations.

All swivel joints and expansion joints shall be 'Plastec' or approved equal and match or exceed the "Ys value" and shall be set at the mid-way point.

Where penetrations of beams and perimeter strip footings are necessary closed-cell polyethylene lagging shall be used around all stormwater and sanitary drainage penetrations through footings. The lagging shall be a minimum of:

- 20 mm thick on Class A to H1 sites
- 40 mm thick on Class H2 and Class E sites.

6.14 Expansion Joints

All piping systems shall be provided with an allowance for expansion. Expansion allowance within pipework may take the form of loops, offsets, expansion joints, fittings or bellows at calculated intervals to prevent stress in pipe work and shall conform to the manufacturer's recommendations and AS/NZS 3500 & AS/NZS 5601. Details for expansion are not necessary shown on drawings, the sub-contractor shall make allowance for correct installation of required expansion joints.

6.15 Building Expansion Joints

Crossing building expansion joints with pipelines shall be avoided where possible. Where this is not possible, lines adjacent to a building expansion joint shall be located in a readily accessible position and fitted with an approved expansion joint, that shall exceed the building expansion joint movement.

6.16 Termite Visual Inspection Zone

Services and items (pipework, hotwater units, downpipes etc) adjacent to buildings, shall be separated from the building by a gap of 25mm or greater to allow clear and uninterrupted visual inspection zone to AS 3660.1 requirements.

6.17 Seismic Restraint

The contractor or their preferred fixings supplier shall produce a compliant seismic bracing design and provide relevant certification, provide equipment and equipment installation suitable for seismic restraint as required by NCC/BCA Sections BP1.1 and B1.2. The reference standard is AS1170.4 Earthquake actions in Australia, and the most relevant section is Section 8 Design of Parts and Components.
For this specification the seismic restraint pertains in particular to:

- Supports & fixings for piping distribution & fire suppression systems
- Fixings and fasteners, especially fixings into concrete and masonry

- Heavy equipment, especially where installed at high level relative to ground level, where installed overhead and where installed in ceiling spaces
- Pipe runs subject to differential movement of building elements
- Fixings for electrical switchboards, data racks, battery systems and the like
- Equipment associated with safety systems and smoke control systems
- Equipment and plant mounted on vibration isolating mounts
- Cable trays and ladders and the attached cables, especially long runs where differential building movement could cause cable damage

Ensure to provide design and installation certification to the satisfaction of the building certifier.

Note: Ashburner Francis are not qualified to provide structural engineering advice and cannot provide any additional information or specific details with respect to seismic restraints.

6.18 Piping Support

Provide proprietary support systems of galvanized or zinc-coated steel construction in non-corrosive areas or of stainless-steel construction in wet or corrosive environments.

Provide anchors and guides to maintain long pipes in position, and supports to balance the mass of the pipe and its contents complete with noggins and supports as required.

Note: Where fire hydrant pipework is likely to be exposed to fire in a building that is not protected by fire sprinklers, then hydrant pipework supports shall have a FRL not less than 60/-/- in accordance with AS 1530.4 and shall be equal to “FRS” (Fire Rated Solutions)

Pipework and fittings shall be:

- Only fixed in approved locations.
- Adequately secured to the structure to support the pipework under full load conditions with a safety margin of 2:1. And meet compliance with seismic restraints as required.
- Kept clear of structure and other services.
- Provided with 0.75mm galvanised sheet metal sleeves where passing through vertical structure. Space between pipe and sleeve to be packed with fire rated material equal to fire rating of structure through which pipe passes or wrap with a flexible material on not less than 20mm thick through horizontal slabs. Ensure termite protection is not compromised.
- Installed to allow for adequate expansion and contraction without causing stress on pipes or joints.
- Fixed on hanger brackets to allow adjustment for fall.
- Join dissimilar metals using fittings of electrolytically compatible material.
- Not be fixed with explosive power tools.
- Cleaned of all cement dropping on completion.
- All fittings / connections to roof purling are to be made to the purling web & approved by the structural engineer.

Materials:

- **Ensure Rubber Barrier / isolation strip** is installed between all pressurised pipes and all brackets, equal to Uni-Cushion.
- Use P.V.C. coated brackets on P.V.C. pipes.
- Use Stainless Steel or Galvanised (to suit area of installation) bolts and fixings of adequate size, to suit the area of installation.
- All mild steel must be hot dipped galvanised, and any cut ends must be painted with galvanised paint.
- Use patented masonry anchors for fixing into masonry elements.
- Pipe clips in timber roof truss spaces shall be PVC coated metal saddle clips.
- Pipe clips shall be equal to Abey / Ezystrut / FRS or approved equal.
- Where supporting fire hydrant pipework, that is likely to be exposed to fire in a building that is not protected by fire sprinklers, then hydrant pipework supports shall have a FRL not less than 60/-/- in accordance with AS 1530.4 and shall be equal to “FRS” (Fire Rated Solutions)
- Shall be in compliance with AS 1170.4, where required for seismic restraint.

NOTE: Hoop Iron or similar strapping will NOT be acceptable for use as pipework supports.

Spacing: Fix piping at the following maximum intervals.

Non Pressure Lines - uPVC – Drainage

Size	Vertical	Horizontal/Graded
40 & 50 mm	1.5 m	1.0 m
65 to 150 mm	2.0 m	1.2 m

225 mm >	2.5 m	1.5 m

Non Pressure Lines - HDPE – Drainage
Refer to manufactures installation guidelines for details. Fixings (guide brackets and anchor points) and expansions joints are based on the expected temperature difference of discharge

Non Pressure Lines - 'Raupiano Plus' – Drainage
Refer to manufactures installation guidelines for details. Sound-dampening, guiding, security and fixing brackets and anchor points

Pressure Lines – PEX - Water

Size	Vertical	Horizontal/Graded
15mm	1.2 m	0.6 m
20mm	1.2 m	0.6 m

Pressure Lines - Copper pipes – Water & Gas

Size	Spacing
15 mm	1.2 m
20 mm	1.2 m
25mm	1.2m
32mm	1.5m
40mm	1.5m
50mm	1.5m
65mm	1.8m
80mm	1.8m
100mm	2.0m

Pressure Lines. – Fire - Hydrants

Size	Spacing
100 - 150 mm	4m
NOTE: Supports shall be no more than 300mm away from any mechanically jointed couplings or hydrant landing valves	

Pressure Lines - uPVC - Pump-out lines

Size	Vertical	Horizontal/Graded
40 & 50 mm	1.5 m	0.9 m
65 to 100 mm	2.0 m	1.2 m

6.19 Pipework Identification

Identify all pipes in accordance with AS 1345 - Identification of the Contents of Pipe. Refer to previous section for information in relation to labels.

Labels shall be equal to Safetyman pattern positioned on each side of valves, bends and junctions and along the length of the pipeline at maximum 3000mm centres.

Where pipes are installed within ducts or false ceiling areas, additional labels shall be installed so as to be visible at the access panel position.

Provide labels on either side of a wall where that pipe passes through the wall.

Ensure the use of chevrons (direction arrows) are correctly identified to indicate the direction of flow

Where noted: Allow to paint all exposed pipework in accordance with client requirements & identify the contents to AS 1345 .

Apply the first coat immediately after substrate preparation and before contamination of the substrate can occur. Ensure each coat of paint or clear finish is uniform in colour, gloss, thickness and texture, and free of runs, sags, blisters, or other discontinuities. All painting shall be carried out by a licensed painting sub-contractor.

Pipework Colours

Pipe Contents	Pipe Paint Colour
Sanitary plumbing and vents	Black
Potable Hot & Cold water supply	Emerald Green
Fire Water	Signal Red
Gas Services	Beige

6.20 Cold and Hot Water Tap Positions

Locate the cold water tap to the right of, or below, the hot tap.

6.21 Isolation Valves

Ensure isolation valves are provided to each group of fixtures (e.g. Male & Female bathrooms) appliances, fixture, and adjacent to each hose tap. For drawing clarity Isolation valves are not necessarily detailed on the drawings. Ensure to coordinate IV's with builder for location of required Access Panels.

Where ensuites or bathrooms are documented, the isolation valving serving the room(s), maybe located within the TMV cabinet (where installed)

- Allow to provide test gates and isolation valves where necessary to suit construction staging and to allow for satisfactory testing of the hydraulic systems.

6.22 Cover Plates

Where piping emerges from exposed building surfaces, provide cover plates of stainless steel, close fitting and firmly fixed in place.

6.23 Thrust Blocks

The hydraulic contactor shall allow all costs to design and install suitability sized thrust blocks and supports, to meet the thrust bearing loads imparted by resultant thrust force to all pipework and fittings to restrain the internal operating pressures under all conditions.

The sizing of each thrust block and support shall be suitable to meet the particular pipework and fitting materials used and local grounds conditions as found at each required location and shall bear against virgin soil material.

Where required engage a specialised soil engineer to assist in the design.

6.24 Protection of In Ground Metal Piping

Where installed underground all new metallic pipelines, pipe fittings, valves and surface areas of existing branch pipelines of any material joining new pipelines shall be provided with a protective Coating, Sleeve or Tape.

<u>Polyethylene continuous sleeve:</u>

The poly sleeve shall be blue wrap equal to wrapping manufactured by Crevet Industries. The sleeving shall be joined by overlapping the ends by no less then 600mm and sealing the joints by use of a manufacture approved PVC adhesive tape.

The sleeving shall be fitted to affect a smooth, closely contoured envelope with minimum barrel and socket gaps. There should be sufficient slack in contouring to allow the sleeving to follow the profile of the spigot-socket interface to prevent damage during backfill.

Every care shall be taken to prevent damage to the PVC wrapping when placing backfill. Any damage to the PVC wrap before or after backfilling shall be made good by the sub-contractor.

<u>Petrolatum taping (Denso tape)</u>:

All in ground metal pipes smaller than 80mm diameter shall be wrapped with a layer of petrolatum impregnated fully synthetic woven carrier tape.

All clean pipes and fittings shall be primed with a compatible priming paste prior to wrapping of the tape.

All pipes and fittings shall be covered with a minimum overlap of 30 millimetres.

Tape shall be equal in all respects to DENSO.

6.25 Protection of Finished Surfaces

Throughout the project, all finished surfaces shall be adequately protected against damage. Provide suitable material for protection of finished surfaces as required and as may be directed.

On completion, remove protection from finished surfaces and leave surfaces in such a manner that final cleaning of dust only is required.

Scratched or damaged finished surfaces will not be accepted.

6.26 Corrosion Protection

All metallic materials and components shall be suitably protected against corrosion.

Bolts, screws, rivets, etc for external use shall be non-ferrous or stainless steel, or where approved galvanised steel.

Brackets, plinths, rods, etc shall be hot dip galvanised after fabrication.

All steel sheets shall be corrosion protected by means of degreasing, priming and painting as a minimum.

Where the application is in a highly corrosive environment such as in close proximity to the sea, provide 316 stainless steel or non-ferrous metallic components.

6.27 Noise and Vibration

The Hydraulic sub-contractor shall be responsible for the elimination of system noise, vibration and water hammer. Special care shall be taken to avoid transmission of vibration using anti-vibration inserts and special purpose support brackets and frames, water hammer devices to isolate pipes, pumps and equipment.

- Ensure the installation and use of rubber barrier / isolation strips between all pressurised pipes and brackets

Allow to comply in all respects to the Acoustic Consultant’s Report and requirements as directed. The acoustic consultant’s report, where available, will be provided upon request. Where a report is not available, allow to comply with NCC standards.

6.28 Brazing

Where silver brazing of copper pipework is necessary, (Press-fit jointing ‘Viega - Propress’ is preferred) ensure an approved silver brazing alloy containing not less than the following is used:

- 15% silver for water & gas services

Flux shall be as recommended by the brazing alloy manufacture.

6.29 Kinco Nut & Olive

The use of compression style kinco nut & nylon / copper olives as a joining methodology for pipework is not permitted, due to high risk of failure.

6.30 Photographic Record

All pipework installed underground shall be photographed prior to backfilling taking place.

Each section of pipe shall be clearly identified and cross referenced to the ‘As Recorded’ drawings.

This photographic record forms part of the Instruction and Maintenance Manual.

6.31 Existing Services

Where work affects the continuity of existing or completed services, arrange the least inconvenient time of interruption, giving at least five (5) working days notification to the Managing Contractor who shall obtain a suitable close down time and / or give notification to local authorities having jurisdiction over the affected services (e.g. QFD) The work shall be organised to minimise the duration of an interruption.

Requirement: Establish that existing services are required to be connected are those to which the documents indicate connection and that they are of the size and level shown on the drawings. No additional claims will be accepted for rectifying works that have been incorrectly connected as a result of failing to confirm the documented information on site prior to commencing any work.

Cleaning, Flushing and Testing: Existing services to which the new pipe work connects (other than Authority Services) shall be cleaned, flushed out and tested to the equivalent standard of new works to the satisfaction of the local authority prior to their connection.

Should services be made redundant, cap and seal the service off in an approved manner as required by the local Authorities and to the approval of the hydraulic consultant.

Verifying Locations of Existing Services: Check with local authorities and the managing sub-contractor concerning the location of existing services on the site. Be responsible for determining the location of all services underground in the vicinity of the works and take precautions to maintain the integrity of these services.

The location of existing services are to be confirmed by vacuum excavation.

7. COMPLETION

7.1 General

All equipment and systems shall be fully operational and complete, all in accordance with manufacturers recommendations, prior to final testing and commissioning.

7.2 Operation and Maintenance Manuals

Provide Operation and Maintenance Manuals for all systems and components covered by this specification   Include As-Installed documentation for the entire installation in these manuals.

Note: Refer BAC requirements – for exact requirements.

Provide 1 hard (paper) copy of manuals and drawings neatly bound into A4 size hard plastic cover ring binders which are permanently labelled with the Project Name, Contract Description and Sub-Contractor's Name.

Provide electronic copies on two (2) USB 3.0 storage devices.
Set up USB drives generally in accordance with the below file structure (with sub-folders for each scope item on the project), with relevant documents filed in the appropriate sub-folder:

Include the following in each set of manuals:

- Index
- Sub-contractor, supplier and manufacturer contact details.
- Hydraulic Services descriptions. (a brief description of the Coldwater, Hotwater etc)
- Product Brochures and Warranty details.
- Performance Based Solution (PBS) documents.
- Reference to following Fire Engineering
- Maintenance schedules. (eg Fire Hydrants, Fire Hose Reels, TMV's, bleed sumps, backwash valve, hotwater systems etc.)
- Replacement Schedule. (e.g. recommend replacement time frames - for items such as flexible connectors, TLV's, tank liner etc)
- A recommended list of spare parts.
- Authority certificates.
- Installation certificates and commissioning forms.
- Fire & Smoke Penetration certificates and schedule shall be included.
- Balancing Valve flow rates and numbered reference (where installed)
- Pump Flow Rates: Cold Water, Hot Water, Fire Service
- Valve Schedule including backflow certification and registration.
- Thermostatic Mixing Valve commissioning forms.
- Photographs of in ground services.
- CCTV survey video results
- Paper prints of all As-Installed Drawings as follows:
  - Hydraulic drawings in the same size and form as the original contract drawings, modified to reflect all changes which occurred during construction, complete with Diags
  - Fire Service Block plans – (where required)
  - Gas Block plans where over 5 storeys - (where required)
  - Hotwater Circulation Block plan – (where required)
- The title block of each drawing shall be amended to include the Sub-Contractor's name and contact details, Sub-Contractor's drawing number and date of issue.
- Refer to "As Installed Drawings" section, for further details.
- A USB 3.0 or suitable electronic storage device containing electronic copies of all As-Installed drawings & block plans in AutoCAD ".dwg" or REVIT format as applicable, complete with a matching set of PDF drawings and a PDF version of the O&M manual. Ensure to Label the electronic storage device.

If additional reference documents or technical sections are required, include corresponding material into the O & M manual.

Include for a matching hydraulic maintenance manual binder complete with, loose leaf log book pages.

Manuals shall be prepared in A4 size loose leaf format. Binders shall be commercial quality, 4 ring with hard covers, each indexed, divided and titled. Include the following features:

- Number pages consecutively.
- Identify each binder with typed or printed title "OPERATION AND MAINTENANCE MANUAL", to spine. Identify title of project, volume number, volume subject matter, and date of issue.
- Ring size shall be 50 mm maximum, with compressor bars.
- Manufacturers' printed data, including associated diagrams, or typewritten, single-sided on bond paper, in clear concise English.
- Provide durable dividers for each separate element, with typed description of system and major equipment components. Clearly print short titles under laminated plastic tabs.
- Fold drawings to A4 size and accommodate them in the binders so that they may be unfolded without being detached from the rings. Provide with reinforced punched binder tabs.
- Ensure a suitable section is contained within the front of the folder to securely hold and retain the electronic storage device.

Provide a draft set of the Operation and Maintenance Manuals to the Superintendent not less than 2 (two) weeks before the date of practical completion, for the purpose of review and comment.

Submit final sets of the Operation and Maintenance Manuals to the Superintendent not more than 2 weeks after the receipt of comments on the draft from the Superintendent.

Contract payments may not be approved until satisfactory Operation and Maintenance Manuals have been received.

7.3 As Installed Drawings

Submit record drawings of all completed works. Drawings shall indicate dimensions, types and location of equipment, cables, piping and ductwork in relation to permanent site features and other underground services. Show the "as installed" locations of building elements, plant and equipment. Show off-the-grid dimensions where applicable. Include relationship to building structure and other services, and changes made during commissioning and the maintenance period. Include diagrammatic drawings of each system showing piping and wiring, and principal items of equipment.

A set of drawings shall be kept on site and progressively marked up by the Sub-contractor as the works proceed to record the locations, inverts and details of all installed services, equipment and valves. Pipework shall be dimensioned from building walls to the centre of all pipes.

Invert levels, reduced levels and positions of all inspection chambers shall be identified. These measurements, levels and positions shall be transferred onto the sub-contractors as installed drawings.

Floor plans, Diagrammatics and Details shall be prepared by the Sub-contractor on a solid-state USB stick, matching the format of the hydraulic contract documents being either “.dwg” (minimum AutoCAD release 2010) or REVIT format as applicable, and in addition include a set of “PDF” drawings and a PDF copy of the O&M manual.

Indicate all services and equipment. Submit to the Head Sub-contractor for review. Standard of drafting and minimum scales shall be equal to design documents, accurately plotted, showing dimensions of pipes, valves and services.

In addition to the material as noted above, provide 1 set of “As Installed” drawings to the Head Sub-contractor and drawings as required to suit the Local Authority.

7.4 Clean Up

At completion clean all Hydraulic equipment, pipework, valves, risers within pathboxes to as-new condition.

Remove all waste and construction debris on a continuous basis and ensure to remove all debris from the install in concealed spaces.

After installation of all services and before applying for Certificate of Practical Completion allow to undertake cleaning of all pipework, fittings, valves, pathboxes, hydraulic plant, sanitary ware and tapware to as-new condition.

7.5 Flushing of Services

All piping, valves and fittings shall be thoroughly cleaned before installation, removing all scale, burrs and obstructions.

Water from the Authority reticulation system shall be used for flushing purposes. Where this is not available and only non-disinfected water is available for flushing, it shall be disinfected in accordance with AS4809:2003 Appendix D.

7.5.1 Pressure Systems

All piping systems containing potable water including associated tanks, water heaters, vessels and equipment shall be thoroughly flushed out as soon as possible after installation to remove all foreign matter.

On completion of the installation of all pressure systems and before application for Certificate of Practical Completion, allow to flush out all piping installations to clear sand, copper filings and foreign matter from the system and ensure trapped air is removed.

Prior to flushing operation, disconnect all thermostatic mixing valves, filters, screens, ball float valve inlets, backflow preventers and other appliances, which may be subjected to damage due to foreign materials. Ensure that the final installation is totally clear of any foreign matter and that such material does not affect all valves, filters, screens and outlets. Replace any and all damaged items of equipment and make all necessary reconnections. Provide Superintendent with a minimum of 48 hours' notice prior to inspection and flushing procedures.

Where flushing is to be undertaken after the installation of thermostatic mixing or tempering valves, all strainers and thermostatic elements/sensors shall be removed, cleaned and replaced after flushing.
Replace any and all damaged items of equipment, make all necessary reconnections.

Pressure piping shall be cleaned and flushed as many times as necessary until the flushed water runs completely clear. The flushing velocity in any section shall not be less than 0.75m/s.

Potable Water Storage Tanks shall be cleaned and disinfected in accordance with AS 3500.1

7.5.2 Drainage Systems

On completion of the installation of all services and before application for Certificate of Practical Completion, allow to lift all access chamber covers and major access gates, clear outs to surface and inspect gravity lines.

Clear out any debris from lines and flush entire system to ensure clean bore. Replace access chamber covers and access gates after inspection, clearing and flushing.

Provide Superintendent with a minimum of 48 hours' notice prior to inspection and flushing procedures.

- Allow to undertaken CCTV survey of all drainage systems. Refer to 'CCTV survey' for more information.

7.6 CCTV Survey & Jetting

On completion of the installation of all services and after Jetting / flushing and clearing of debris and before application for Certificate of Practical Completion, allow to undertake a CCTV survey and video recordings of all gravity drainage systems. The survey is to utilise the 'As-built drawings' as a reference to each section that has been surveyed.

Where deflects, blockages or other issues are noted within the recordings the contractor shall rectify.

Include all recordings in the O&M manual, on suitable USB drive.

7.7 Testing and Commissioning

7.7.1 General

Fully test and commission the complete hydraulic installation prior to practical completion.

Only suitably qualified personnel shall carry out testing and commissioning. Provide evidence of qualifications when requested.

Record all test results and submit for inspection prior to practical completion.

Carry out functional and operational checks on equipment and circuits and make adjustments for the correct operation of safety devices.

Undertake necessary pressure testing, calibration of all pressure services, equipment and valves.

Securely anchor pipes and fittings in position to prevent movement during tests. Leave pipe joints exposed to enable observation during tests.

Testing should not commence before the curing time for any installed concrete thrust block has elapsed.

Disconnect and isolate equipment which is not designed to carry the test pressure.

Submit reports indicating observations and results of tests and compliance or non-compliance with requirements. Give sufficient notice for inspection to be made of the commissioning of the installation.

Hot water plants shall be tested and commissioned by the water heater manufacturer. Obtain a clearance installation certificate from the manufacturer.

All testing is to be carried out prior to pipework being back-filled with joints clearly visible for the duration of the test.

The below testing shall be the minimum general testing requirements. Where local authority or Australian Standards testing requirements exceed the below parameters, the contactor shall allow to undertake and ensure compliance with the higher of the testing requirements.

7.7.2 Water Services Tests

The water service test shall be carried out according to the methods and requirements set out in AS/NZS3500.1 for all water services.

Generally, the water service test shall be a hydrostatic test to 1500kPa for a period not less than 30 minutes with no leakage. The test is to be carried out prior to pipework being back-filled with joints clearly visible for the duration of the test.

The Sub-Contractor shall ensure that any equipment which may be damaged by the test pressure is disconnected and pipework capped prior to the test being carried out.

The whole water system including valves, pumps and all other equipment shall be operated under normal conditions for a period not less than 48 hours without fault to be deemed operational.

7.7.3 Heated Water Services Tests

The water service test shall be carried out according to the methods and requirements set out in AS/NZS3500.4

Generally, the water service test shall be a hydrostatic test to 1500kPa for a period not less than 30 minutes with no leakage. The test is to be carried out prior to pipework being back-filled, insulated or concealed in ducts, chases or trenches, with joints clearly visible for the duration of the test.

The Sub-Contractor shall ensure that any equipment which may be damaged by the test pressure is disconnected and pipework capped prior to the test being carried out.

The whole heated water system including valves, pumps, water heaters and all other equipment shall be operated under normal conditions for a period not less than 48 hours without fault to be deemed operational.

Where balancing valves are installed the hydraulic sub-contractor shall allow all costs to commission and balance the flow and return system to the satisfaction of the Hydraulic Consultant. (Where required, allow to engage the use of a specialised contractor to balance the system).

Allow to clearly number and record all balancing flow rates within the O&M Manual against each balancing valve and include for an additional electronic copy of the hotwater circulation diagram to be included within the O&M manual.

Refer to AS 3500.4 for additional requirements & the hotwater section of this specification.

7.7.4 Sanitary Plumbing & Drainage Tests

The sanitary plumbing and drainage test shall be carried out according to the methods and requirements set out in AS/NZS3500.2

Generally, the Sanitary Plumbing test shall be a hydrostatic test to flood level for duration of no less than 15 minutes or, air test to 15 kPa for duration of no less than 3 minutes to allow the air pressure to stabilise. Then reduce the air pressure to 10 kPa and being recording the time and drop in pressure against Table 15.3.2. There shall not be a pressure drop greater than 3 kPa.

Generally, the Sanitary Drainage test shall be the hydrostatic water test to 1 meter above the soffit level at the highest pint of test section for duration of no less than 15 minutes Alternatively, the air test shall consist of applying a pressure of 15 kPa for duration of no less than 3 minutes to allow the air pressure to stabilise. Then reduce the air pressure to 10 kPa and being recording the time and drop in pressure against Table 15.3.2. There shall not be a pressure drop greater than 3 kPa.

Where Vacuum Drainage and Plumbing is utilised, a vacuum test shall be undertaken in accordance AS/NZS3500.2 and shall meet the manufactures requirements.

7.7.5 Stormwater Drainage and Downpipe Tests

The stormwater drainage test shall be carried out according to the methods and requirements set out in AS/NZS3500.3

Generally, the downpipes shall be subjected to a hydrostatic test to the lesser of 10 metres head or head of water equal to the length of the downpipe for a period of not less than 10 min. Alternatively, air test to 30kPa for a duration no less than 3 minutes.

Generally, the stormwater drainage test shall be a hydrostatic test to 3 metres of water for a period no less than 10 minutes. Alternatively, the air test shall consist of applying a pressure of 30kPa to the drain for 3 minutes. The air supply shall then be shut off and the time measured for the pressure to drop from 25kPa to 20kPa.

The test is to be carried out prior to pipework being back-filled with joints clearly visible for the duration of the test.

7.7.6 Pump Out and Riser Main Tests

Pump Out and Rising Mains shall be free of leaks when subjected to a pressure of no less than twice the shut-off head of the pump connected to the pipework, for a period of not less than 10 min.

7.7.7 Fire Hydrant System Tests

The fire hydrant system test shall be carried out according to the methods and requirements set out in AS2419.1, AS 1851 and the local fire bridge. (QFD & ARFF)

Upon completion, the Sub-Contractor shall remove any air trapped within the system and flush the system to remove any debris that may have accumulated within.

At the elevation of the highest fire hydrant within each pressure zone, the system shall be pressurised to not less than 1.5 times the highest pressure or 1700kPa, whichever is greater.

Pipework shall be tested hydrostatically tested to the greater of 1700 kPa in accordance with 'Tests 1 to 4' in appendix S of AS 2419.1 – 2021.

The test shall be applied for a duration not less than 2 hours.

Every hydrant shall be opened to confirm the presence and flow of water.

In addition, discharge the required number of the most hydraulically disadvantaged hydrants, each at not less than the required outlet pressure to confirm the required flow rate.

<u>Internal Fire Hydrant Test Drain Facilities</u>

Ensure internal fire hydrant test drains (where required) are installed to meet AS 2419.1-2021 complete with test facility as per <u>figure N.9.2</u> at the most disadvantaged hydrant(s) and shall include:

- Flow test tubes (or annubars) that allows for a differential flow sensing elements to be fitted. Flow test tubes of 100mm and larger shall be provided with two (2) element sockets installed as per <u>figure N.9.2.1</u>
- Valving to allow for the isolation of the flow test tube and the connection, disconnection and operation of the test gauges and flow sensing element
- Permanent test drain(s)
- Permanently connected hose fittings in the test drain that allow for the fire hydrant valve to be directly connected to this fitting through the use of firefighting hose, complete with cap and chain
- Labelling and each permanent hose fitting is provided with their own anti-tamper cap and chain.

Ensure each Fire Hydrant has been provided with a cap and chain and a suitable testing record tag has been attached indicating the month and year of commissioning.

<u>Fire Tanks – Reduced-capacity tanks test facility</u>

Where a reduced-capacity fire tank is installed, ensure a permanent flow test facility is installed at the automatic infill supply pipework to the tank(s) to meet AS 2419.1-2021 and shall include:

- An element socket located less than 2.4m above the ground or floor level at which a residual pressure gauge and flow sensing flow element is able to be connected
- Include valving to allow the connection, disconnection and operation of the test gauge and flow sensing element; and
- Be sized to measure the automatic infill rate as shown on the block plan

Note: Where automatic diaphragm (pilot operated) valves are installed external to the tank(s), a 15mm drain valve(s) shall be installed to operate the valve, so the flow can be initiated during the flow testing from ground level.

Complete and submit all necessary Authority forms (including within QLD a ‘Form 71’) to obtain a compliance certificate from the Fire Authority. Retain a copy of the certificate within the O & M Manual.

Ensure fire block plans are provided, hydrants are numbered and meet the satisfaction of the local fire authority before being permanently manufactured and installed.

Allow to engage a specialised sub-contractor to undertake the testing as required.

7.7.8 Fire Hose Reel Tests

The fire hydrant system test shall be carried out according to the methods and requirements set out in AS 2441, AS 1851 and the local fire bridge.

Upon completion, the Sub-Contractor shall remove any air trapped within the system and flush the system to remove any debris that may have accumulated within.

The Sub-Contractor shall confirm the following: -

- That there is no leakage from the assembly when the hose is charged with water and nozzle is closed.
- That the hose can be easily unwound in any direction.
- That the flow rate through the nozzle is as stipulated in AS 2441
- That any defects have been addressed.
- That the hose can be retracted back in even layers, with the stop valve open.
- That each fire hose reel is left depressurized with the valve in the closed position.
- A suitable testing tag to AS 1851, has been provided to each FHR indicating the month and year of commissioning.

Allow to engage a specialised sub-contractor to undertake the testing as required.

7.7.9 Backflow Prevention Device Tests

All testable backflow prevention devices shall be tested and commissioned by a licensed tester. The Sub-Contractor shall arrange for and bear all fees and charges associated with such tests.

The results of tests carried out on backflow prevention devices shall be recorded and the commissioning results shall be included in each of the Operation and Maintenance manuals.

Allow to register on behalf of the client any required backflow prevention devices in accordance with the local authority.

7.7.10 Valves, Fixtures and Fittings Tests

All valves (isolating valves, check valves, tempering valves, reflux valves, pressure reduction valves etc.) and fixtures & fittings shall be tested and commissioned to ensure they are functioning in the manner intended.

All filters and strainers shall be cleaned and reinstalled with any faulty items being replaced and re-tested to ensure correct operation.

7.7.11 Pump Tests

All pumps and associated control panels shall be tested and commissioned to meet the operational requirements of the project and services being delivered.

Allow to have these items commissioned by the supplier, complete with written confirmation that the pump(s) meet the operational requirements and are watermark certified where required.

7.8 Greenstar Testing and Commissioning Requirements

Carry out testing and commissioning in accordance with Green Star requirements, including CIBSE Commissioning Codes.

After practical completion, complete quarterly commissioning reviews and a final recommissioning after 12 months of operation.

After each commissioning period, provide a report which confirms the commissioning of the system is in compliance with contract documents. The report shall also include a documented list of outstanding commissioning issues, records of all functional / commissioning testing undertaken and list any seasonal testing required in the future.

7.9 Test Instrument Calibration

All test instruments shall have current calibration certificates. Certificate copies shall be provided to the Superintendent prior to commencement of tests.

7.10 Applications

Coordinate the actions required by all parties including owner, tenants, utility retailers and metering providers to ensure the hydraulic services are installed to meet project timeframes.

7.11 Inspection at Practical Completion

Arrange for a Practical Completion Inspection by the Architect. A minimum of 5 working days' notice shall be given.

Ensure that the installation is fully operable and practically completed before requesting an inspection.

Should the inspection reveal that the installation is not fully operable or practically completed and require further re-inspections to complete the process, these re-inspections will be charged to the Hydraulic Sub-contractor.

7.12 Practical Completion

Practical Completion will be granted after the following is provided:

- Finalisation of all testing and commissioning
- Certification that the installation complies with the specified requirements and all applicable Statutes and Standards
- All authority approvals are obtained as required.
- Instruction on safe plant and equipment operation has been provided to the users.
- Operation and Maintenance Manuals and As-Installed drawings have been provided.
- Defects have been rectified.
- When project or site-specific contractual requirements have been met.

7.13 Instructions and Training

Provide skilled operators to instruct such persons as may be nominated by the Architect in the effective operation of all of the hydraulic systems in the installation.

Instructions shall not be given until completion of all tests, commissioning and adjustments have been made.

Conduct training at agreed times and locations and for durations commensurate with the complexity of the systems. Use items and procedures listed in the operation and maintenance manuals as the basis for instruction. Review contents with the principal's staff in detail and amend as required on completion of training.

7.14 Maintenance and Warranty During Defects Liability Period

Carry out preventative maintenance on a regular schedule for systems as detailed in other sections of this specification.

Promptly rectify all defects during the defects liability period.

Allow all costs within the tender to maintain all equipment, fixtures, backflow valves, thermostatic mixing valves etc, for a minimum period as the contract dictates or as a minimum period of 12 months.

During the maintenance period, carry out periodic inspections and maintenance work as recommended by manufacturers of supplied equipment and as required by Australian Standards.

Attend emergency calls promptly during this period. On completion of the maintenance period advise the owner the period has ceased and provide all necessary maintenance forms.

Submit details of maintenance procedures and program, relating to installed plant and equipment, 6 weeks before the date for practical completion. Indicate dates of service visits. State contact telephone numbers of service operators and describe arrangements for emergency calls.

Report to the principal's designated representative on arriving at and before leaving the site while undertaking maintenance procedures.

Submit in binders which match the O&M manuals, loose leaf log book pages designed for recording completion activities including operational and maintenance procedures, materials used, test results, comments for future maintenance actions and notes covering the condition of the installation. Include completed log book pages recording the operational and maintenance activities performed up to the time of practical completion.

Provide sufficient pages for the maintenance period and a further 12 months.

Record all maintenance visits and comment on the functioning of the systems, work carried out, items requiring corrective action, adjustments made and name of service operator. Obtain the signature of the principal's designated representative.

If referenced documents or technical sections require that logbooks or records be submitted, include this material in the maintenance records.

On satisfactory completion of the installation, submit certificates stating that each installation is operating correctly.

8. EXCAVATION AND BACKFILLING

8.1 General

This section is applicable to all in-ground services.

For the duration of the contract, the Sub-contractor shall maintain in a well-drained condition all those parts of the site including excavations that from time to time are in his possession. No bedding material or pipework shall be placed in excavations containing standing or running water.

Excavate to the lines, levels and grades as required for underground services specified in the relevant sections. Unless otherwise specified make the trenches straight between manholes, inspection pits, junctions and the like, with vertical side and uniform grades. Make allowance for all shoring, strutting or other necessary ground support.

Paved areas shall be saw cut prior to excavation.

Excavate trenches in sections of suitable length, and after the relevant service length has been laid and bedded, backfill the trench section, with the minimum of delay, and if possible on the same working day.

Keep trench widths to the minimum, consistent with the laying and bedding of the relevant service, and the construction of manholes and pits.

Trench depths shall be as required by the relevant service and its bedding method.

The bottom of each excavation shall be firm and stable.

The thickness of bedding under pipes and fittings shall not be less than 75mm, after compaction.

If in the opinion of the Superintendent or a nominated representative, the conditions causing or requiring over-excavation and backfilling, and consequent additional works have been brought about by negligence or methods of work, no additional compensation will be made.

All excavations shall be subject to inspection before any services installations and construction works are permitted to proceed.

To provide clearance for sockets, where socketed pipes are used, additional depressions shall be excavated in the base of trenches. The pipes shall be installed with full bearing on barrels only, and sockets shall not be bear on ground or bedding.

Service trenches shall be backfilled as soon as possible after approval of laid and bedded service and compacted to the density which applies to the location of the service trench. Compact so that pipe is buttressed by the walls of the trench.

Supporting material shall be removed during the backfilling operation.

Where service excavations occur in topsoil areas, complete the backfilling with topsoil.

All disused excavated material and contract debris shall be removed from site and disposed in a legal place. Allow to pay all costs associated with removal of unwanted spoil.

8.2 Safeguards and Precautions

The operator of any excavator / back-hoe shall be suitability qualified and licensed to operate the machinery.

Excavations deeper than 1.5meters are deemed “high risk construction work” and must be accompanied by a Safe Work Method Statement (SWMS).

The Sub-contractor shall carry out the work in a careful, secure, safe and tidy manner and take all precautions against damage whether arising from bad workmanship, breakage, machinery, plant, insufficient timbering/shoring, flooding or any other cause whatsoever.

Ensure against collapse with adequate shoring; provide all warning lights, barricades, planking etc., as necessary to ensure safe working conditions. If excavating adjacent to structures, determine depth of structure footing, the stability of walls so as not to undermine footings or suffer collapse of walls etc.

Provide, erect and maintain warning signs, temporary fences, barriers and night lights, adjacent to any works such as trenches and excavations or stacks of material which could be considered a danger to persons or traffic of any kind.

Obey all directions given with regard to the provisions of lighting and barriers, but shall not be relieved of the responsibility to provide adequate lighting and barriers to prevent any accident or damage.

8.3 Shoring of Excavation

Subject to any Act of Parliament, Ordinance or Regulation, the Sub-contractor shall satisfy themselves as to the necessity of shoring any excavation and shall accept the sole responsibility as to its being required and to its use in the works.

Where necessary for safe and efficient completion of the work erect shoring and timbering of sufficient strength and quality to prevent earth and other material slipping or falling in or being shaken from the side of the excavation. As the work proceeds, all shoring and timbering shall be withdrawn except in the cases where the Head Sub-contractor has indicated in writing that such shoring and/or timbering shall be left in position. The supply, erection and withdrawal of timber work shall be considered to be included in the cost of excavation.

8.4 Soil Classification and Ground Quality

Prior to commencement on site, obtain a copy of the soil report to get an understanding of the soil classification for the site and take into account any soil conditions likely to affect the in-ground installation and extent of this contract.

Upon commencement of excavations a general examination of the trenches shall be made to determine the presence of foreign material, fill or water charged ground. If the sub-contractor is unsure of the material encountered advice shall be obtained from a qualified geotechnical engineer prior to commencing of any pipe installations.

If foreign, fill or water charged material is encountered, rectification work of the ground shall be carried out to ensure the trench base has adequate bearing capacity to carry any possible loads placed on both the trench and pipeline, avoiding any possible future subsidence. The sub-contractor shall obtain from a qualified geotechnical engineer the required remedial action and advise the Head Contractor with the recommendation and costs of remedial action to be taken. No remedial work is to commence until written approval to proceed is obtained from the Head Contractor.

Should the hydraulic sub-contractor ignore or fail to advise the Head Contractor of any foreign material encountered prior to laying of pipelines, shall render the hydraulic sub-contractor liable for any possible rectification works.

8.5 Existing Services

Care shall be taken to avoid damage to existing underground services. Excavation in the areas of known services locations shall be carried out by vacuum truck excavation.

Care shall be taken to protect installed services located in ground. Clearly identify the location of installed drain lines on site and make the managing contractor and other sub-contractors aware of their location.

8.6 Boring

Where required by the relevant authorities, or as noted on drawings, provide boring, by an approved specialist contractor, in lieu of trenches.
Make the bored dimension slightly less than the relevant service pipe to ensure a tight fit. If voids are encountered, or bore oversized, fill by pressure grouting with not less than 3% stabilised sand.

8.7 Excavation

All trenches shall be excavated straight with the appropriate uniform grades between pits and changes in direction.

Where excavations extend through existing pavements, the pavements shall be saw-cut at a distance of 200mm from each side of the trench and re-instated, or alternatively bore under paving for pipe installation.

Where excavations extend through lawn, the side of the trench shall be defined by cutting through the turf. The turf shall be cut into sods and together with all topsoil under them (to 150mm overall depth) shall be removed, stacked and nurtured for re-use.

For all excavations of trenches beyond the limits of all paved and lawn areas the excavations shall be carried out so that the initial topsoil is placed to one side while the sub-soil is placed to the other.

Excess or unsuitable materials from trench excavation shall be removed from the site and disposed of in a legal place.

8.8 Trench Widths

Keep trench widths to the minimum consistent width required for the laying and bedding of services and pits.

This does not omit the subcontractor from his obligations in providing a safe trench width in accordance with any workplace health & safety act. If required widen trenches where required by this act.

8.9 Trench Depths

If excavation is necessary below the level of any footing or footings, seek approval from the structural engineer prior to excavating any trench located within close proximity to these footings. If necessary provide additional support for the footings.

Advise the Head Contractor of the additional footing requirements prior to any excavations.

8.10 Obstructions

Keep trenches clear of sharp objects. If necessary cut back roots to at least 600 mm clear of services. Remove other obstructions including stumps and boulders, which interfere with services or bedding.

If rock is encountered, give notice. Additional payment for rock in trench excavation shall only be considered where material cannot be removed by a 12 tonne hydraulic excavator fitted with a single tooth ripper and/or tiger teeth or boulders with a min. dimension of 1m.

Blasting and use of explosives are NOT permitted.

8.11 De-watering

Generally keep trenches free of water. Place bedding material, services and backfilling on firm ground free of surface water. If dewatering is required advise the Head Contractor with costs to cover dewatering of trenches, pits, manholes, etc. prior to carrying out any dewatering.
Where water is present, the level shall be lowered below the bottom of the trench and maintained at that level during construction and including during the placing of trench fill.

8.12 Excess Excavation

If the Hydraulic Services Sub-contractor has exceeded the section area of excavation, as shown, in consequence of any injudicious working, slips, falls, or for any cause other than by direction of the Superintendent then the Hydraulic Services Sub-contractor shall, at sole cost, remove such extra materials from the site.

The excess depth of over-excavated trenches shall be filled with bedding material compacted to achieve a density as near as possible to the original soil density and to the satisfaction of the Superintendent.

Where unsuitable material is encountered at the base of excavations, obtain the Superintendent's prior approval before excavating and making good as directed and/or specified.

The Superintendent will make no extra payment for excavation in excess of that required by the drawings and specification unless directed in writing.

8.13 Bedding, Side Support & Overlay

Where trench base is of stable sand material, pipe barrels shall be directly supported on the undisturbed natural ground.

Generally, all pipelines in-ground shall be supported by the following (minimum) pipework supports consisting of:

- Bedding Base 75mm
- Side Supports 100mm
- Pipe Overlay 75mm

The **materials** shall be made of either:

- **Compacted free-running sand capable of passing through a 2mm mesh sieve**, that does not contain clay, organic or any other deleterious materials.

  or

- **Compacted fine grained gravel capable of passing through a 2mm- 10mm mesh sieve** with no hard edged / sharp objects in contact with or resting against any pipe or fitting and **shall be free from rock or rubble, hard matter or organic material**. (Crusher Dust is not an acceptable material).

Pipe side support and pipe overlay shall not be inferior to the pipe bedding material.
For further detailed information reference shall be made to the applicable parts within AS 3500, namely parts 1, 2 & 3.

Where a mechanical vibrator is employed, the lines shall be subjected to a second test upon completion of the compaction.

8.14 Backfilling

Do not backfill excavations until relevant services installation works have been inspected, tested & approved. Backfill material shall not be dropped from a height of more than 1 metre unrestricted fall.

All backfilling material shall be free of builder's waste, bricks, concrete pieces, rubbish, topsoil, organic matter and lumps of clay or rock with a dimension in excess of 75mm.

Particular care shall be taken to ensure that the material is fully packed under and at both sides of the pipe drain, and that no pipes, fittings or joints are damaged in any way.

Use underground warning tape to identify each in-ground service. warning tape shall be placed 150mm above the service it identifies and of the appropriate wording to correctly identify the service and shall be equal to Tapex Fair Warning Non Detectable PVC Tape.

Unless instructed otherwise where a length of trench has been inspected, the bedding and backfill materials shall be placed and compacted, as soon as possible.

8.15 Compaction

All trenches shall be compacted to achieve a minimum of 95% density when tested in accordance with Standard Compaction methods and in accordance with AS 1289 - Methods of Testing Soil for Engineering Purposes and as follows;

After sand has been placed within the trench, the trench shall be compacted using water to flood the sand to provide compaction.

Selected excavated material / backfill shall be compacted in 500mm layers above the 'overlay layer' up to the underside of the finished working surface. To achieve the required compaction the use of upright ramming machinery, sheep foots rollers or other approved methods must be used. Tests shall be carried out on each 500mm of compacted backfill within each 10metres of trench length to ensure the appropriate amount of moisture content is added and the appropriate passes using the compaction equipment is achieving the required compaction results.

If water charged ground is encountered, gravel shall be used as backfill material to a level above the water charged ground. Geotechnical material shall then be placed over the entire gravel backfill and selected fill material shall then be placed over the top of the geotechnical material and compacted as noted above. The gravel itself does not require compaction or testing

8.15.1 Compaction under buildings, Road & Pavements

With reference being made to the structural and civil engineers' requirements and unless noted otherwise, trenches under buildings, roads and pavements shall be compacted to achieve a minimum of 98% density when tested in accordance with Standard Compaction methods and in accordance with AS 1289 - Methods of Testing Soil for Engineering Purposes.

9. SANITARY PLUMBING AND DRAINAGE

9.1 General

Comply with the General Hydraulic Services Requirements. The system includes for connection to the authority connection provided and extended throughout the building. This includes for in ground drainage, elevated drainage, stack systems, plumbing and associated vents.

Supply, install, test and commission the sanitary drainage services to the requirements of Nation Construction Code of Australia, Plumbing Code of Australia, AS/NZS 3500.2, relevant Codes, the local Authority and to the satisfaction of Superintendent and hydraulic consultant.

9.2 Inspections

Give sufficient notice so that inspections may be made at the following minimum stages:

- Excavated surfaces
- Installation of pipework
- Testing of installation
- Cleaning of pipework at completion

9.3 Pipework and Materials

Refer to “Materials” section for details.

All materials used shall comply with SAA MP52 and be water marked, unless otherwise required by the statutory authority.

9.4 Testing

Refer to “Completion” section for details.

Provide a hydrostatic or air test to choke level for a minimum period of 15 minutes. Complete certification forms for each test carried out and submit to the Hydraulic Consultant.

9.5 Sanitary Plumbing and Drainage

Supply and install all sanitary plumbing and drainage from fixtures to the sewer connection indicated on drawings. Provide all necessary pipes, stacks, vents, junctions, bends, floor wastes, excavation, supports, backfilling, testing and all sundry equipment required for the installation.

Co-ordination with all other disciplines shall be carried out prior to any setting out, excavation and pipe installation taking place.

Execute the works, using only materials and structures as approved by the local Authority.

Supply and install inspection openings as indicated on drawings. Inspection openings shall be a PVC-U riser and incorporate a stainless steel trap screw when installed external of the building and chrome plated finish when installed internally and of the clamp down type where in vinyl areas.

Where installed internally all drainage services shall be installed in a position to maintain maximum available head height. Note: A maximum head height to basement and car parking levels shall be maintained in accordance with the authority (local council) and Australian Standard requirements.

Ensure to comply with and allow all costs for local authorities’ requirements, such as the addition of an sampling points / manholes as required. (e.g. GCCC – requires the inclusion of a boundary manhole to all commercial sites, to allow for future sewer sampling to be undertaken)

9.6 Invert Levels

Liaise with the civil contractor, utility provider and where required conduct a survey to confirm the depth, position and suitability of the sewer drainage connection point before commencing any work and ensure that the new sanitary drainage can gravitate to the proposed connection point.

All proposed invert levels on the drawings are recommended only and are based on information available at the time of design. Invert levels must be checked on site before excavation or installation of pipe work to ensure drainage can gravitate to the relevant connection points and correct cover and gradient can be achieved.

The Superintendent shall be notified immediately of any changes required in order to complete the work. Redundant work resulting from failure to comply with this requirement will void any claims made.

No work is to commence until this has been completed.

NOTE:
In accordance with AS 3500.2:

- Where an 'equal 45° junction' is used to make a connection to a branch drainage line, the entry level of the branch drain shall be rotated at the junction with an incline of not less than 15° above the horizontal. The invert of the branch drain shall at least 70mm higher than graded drain to which it connects.

- Where an 'unequal sized 45° junction' is used to make a connection to a branch drainage line, the invert of the branch drain shall be rotated at the junction to ensure the invert of the unequal branch is at least 10mm higher than the soffit of the drain to which it connects.

9.7 Vents

Terminate vents through the roof at a minimum height of 150mm or as per AS 3500.2.2. requirements. Vent pipes above roof(s) shall be PVC-U DWV grade. Provide approved type vent cowls, of material to match the vent pipes.

Each vent penetration shall be flashed with a "Dektite" made of a flexible and UV resistant EPDM, as manufactured by Deks and shall be installed according to manufacturer's instructions.

The vent shall be painted to match the roof colour where require. (refer to Architects documentation)

9.7.1 Air Admittance Valves

Where indicated on the drawings, Air Admittance Valves (AAV's) complying with requirements of AS3500.2 and AS4936 may be utilised where indicated on the drawings.

They shall not be used for upstream venting of a main drain, used as stacks vents where greater than 10 levels, nor installed as RVA (Reduced Velocity Stack) stack vents.

All AAV's shall be equal to Studor and shall be installed in a readily accessible location, allowing for servicing and or replacement.

AAV's shall not be used unless clearly shown on design documentation or approved hydraulic consultant.

9.8 Sanitary Fixtures

Refer to architectural documentation for the sanitary fixture selection.

Allow all costs to install all sanitary fixtures to the manufacturer's instructions ensuring that the installation is complete with all necessary supports, fixings and the like. All fixtures are to be installed securely and neatly into position.

Coordinate with other trades to provide cut outs in bench tops and hole preparation as required. Allow to provide technical details and samples of fixtures to enable others to perform cut-outs.

All Fixtures, appliances and brackets shall be first quality.

All exposed brackets shall be white enamelled, unless noted otherwise.

All exposed connections shall be chrome plated unless specified in white PVC or otherwise.

Obtain a written guarantee from the manufacturer for a period of two years. The guarantee shall cover all costs associated with replacement of a faulty article within a two year period.

9.9 Fixture Traps

Fixture traps shall be universal traps of approved manufacture and of equal material to which their discharge connects.

Chrome plated copper where basins or sinks are exposed

Traps shall be 50 mm diameter for sinks.

Bath Traps:
Where a bath trap is not accessible, the bath trap shall discharge untrapped to a floor waste gully (FWG) in accordance with AS 3500.2.2021 section 13.9 & appendix B (ie 1.2m untrapped)

9.10 Reducing fittings

All reducing fittings installed within the drainage system are to be eccentric taper fittings / Level Invert Taper (LIT) type and shall be installed obvert to obvert to allow maximum air movement.

9.11 Inspection Openings & Expansion Joints

Expansion joints & inspection openings are to be located on all new services to the requirements of local authorities and manufactures recommendations.

Additional openings shall be provided to facilitate flushing of whole system and for testing purposes as required.

All inspection openings shall be located such that they are readily accessible and where located externally are provided with in-ground expansion joints to suit the expected ground movement.

9.12 Overflow Relief Gully

An overflow relief gully shall be provided to the site. It shall be located in the position as indicated on the hydraulic services drawings. The gully shall be installed 150mm below the lowest fixture outlet in the building and 75mm above the surrounding natural ground level.

This height can be reduced if the surrounding ground is concrete or paved where it shall be finished at a level so as to prevent the ponding and ingress of water.

Overflow relief gullies shall be charged by either a fixture or a hose tap.

Ensure expansion joints are provided to the ORG riser pipework in-ground to suit expected ground movement.

9.13 Reflux Valves

Provide reflux valves where indicated on the drawings and as required to protect sanitary drainage from surcharging in scenarios such as where:

- ORG minimum heights cannot be achieved
- Declared Flood Levels (DFL's) prevent ORG installations
- Authorities deemed such a device shall be installed

Refer to "Materials" section for details.

9.14 Concrete Support Pads

Where drainage services are located in-ground, ensure to provide a minimum 100mm thick concrete (25mpa) support pad located below all sanitary and trade waste drainage turn-ups, gullies, floor wastes, clear outs, access openings, inspection openings.

All concrete pads are to be visible for inspection prior to backfilling.

9.15 Floor Waste Gullies

Floor wastes shall be 100mm PVC-U trap of self-cleansing pattern as indicated on the drawings.

For all floor wastes located within tiled area provide a water proof flanged "Wondercap" within the riser to avoid cement droppings from entering the trap.

Generally, floor waste gullies shall be of type, which matches the floor finish as selected by the architect. (eg Vinyl, Tile, Concrete, Timber finish etc) The grate and surround shall be chrome plated in finished areas.

Floor waste gullies shall be located at the lowest point of the floor within the room. The Sub-contractor shall ensure that the grate and surround are flush with the floor in order to facilitate the effective drainage of spills.

Where located in-ground, ensure to provide a minimum 100mm thick concrete (25mpa) support pad located below.

Where commercial laundry tubs and/or commercial washing machines discharge to a floor waste, ensure to provide anti-foaming devices within FWG's.

Refer to architectural documentation for the sanitary fixture selection.

9.16 Plug and wastes

All plug & wastes installed within sinks and basins shall be of brass chrome plated construction, unless noted otherwise.

9.17 Pan Collars

Unless otherwise noted, all pan collars to soil type fixtures shall be of approved PVC-U material. Collar shall incorporate an approved neoprene rubber ring joint.

9.18 Tundishes

Supply and install Chrome Plated copper tundishes made from 1.2mm sheet copper and finished with rolled edges. Tundishes shall be connected to a threaded outlet of 20mm unless indicated otherwise on drawings and fitted to universal "P" or "S" traps where indicated.

Chrome plate where positioned in finished areas.

Ensure that tundishes are installed in a visible and accessible location with physical air gaps as stipulated in AS/NZS 3500.2. 25mm min & twice the size of the pipe discharging over the TD – whichever is the greater

9.19 In-wall Tundish

Supply and install in-wall tundishes complete with integrated convex base, removable stainless steel cover plate and viewing window.

- "HP Tundish" by 'Ruby Industries'

Where noted, in-wall tundishes shall be trapped via a 40mm 'HepVo' self-sealing waterless trap.

Ensure that tundishes are installed in a visible and accessible location with physical air gaps as stipulated in AS/NZS 3500.2. 25mm min & twice the size of the pipe discharging over the TD – whichever is the greater

9.20 Acoustic Requirements

Where sanitary plumbing and drainage is located within any habitable and non-Habitable spaces acoustic rated insulation shall be installed to meet NCC requirements.

Refer to "Materials" acoustic lagging section for details.

Note: Where an acoustic report has been provided, the hydraulic contractor shall comply with the acoustic performance requirements stipulated in the acoustic report.

9.21 Reactive Soil & Expansion Joints

Where drainage is being installed within reactive soils (Class M, H1, H2 & E) the hydraulic services sub-contractor shall install swivel joints and expansion joints in accordance with the structural engineers' recommendations & in accordance with the Hydraulic Details.

All swivel joints and expansion joints shall be 'Plastec' or approved equal, and match or exceed the Ys value and shall be set at the mid-way point.

9.22 CCTV Survey & Jetting

On completion of the installation of all services and after Jetting / flushing and clearing of debris and before application for Certificate of Practical Completion, allow to undertake a CCTV survey and video recordings of all gravity drainage systems. The survey is to utilise the 'As-built drawings' as a reference to each section that has been surveyed.

Where deflects, blockages or other issues are noted within the recordings the contractor shall rectify.

Include all recordings in the O&M manual, on suitable USB drive.

10. TRADE WASTER PLUMBING AND DRAINAGE

10.1 General

Comply with the General Hydraulic Services Requirements. The system includes for connection to the authority connection provided and extended throughout the building. This includes for in ground tradewaste drainage, elevated drainage, stack systems, expansion joints, plumbing and associated vents.

Supply, install, test and commission the tradewaste drainage services to the requirements of Nation Construction Code of Australia, Plumbing Code of Australia, AS/NZS 3500.2, relevant Codes, the local Authority, utility provider and to the satisfaction of Superintendent and hydraulic consultant.

Ensure to comply with the local authorities' anomalies, such as the addition of extra COS points or manholes for sampling within the drainage system where required.

10.2 Inspections

Give sufficient notice so that inspections may be made at the following minimum stages:

- Excavated surfaces
- Installation of pipework
- Testing of installation
- Cleaning of pipework at completion

10.3 Pipework and Materials

Refer to "Materials" section for details.

All materials used shall comply with SAA MP52 and be water marked, unless otherwise required by the statutory authority.

10.4 Authorised Trade Waste Products

It is the responsibility of the hydraulic sub-contractor to confirm all trade waste pre-treatment devices or products that are specified or utilised within the project are approved for use by the local trade waste authority.

The approval confirmation process is to occur prior to ordering of the device. Where the device is found to be un-approved for the local area, the hydraulic sub-contractor shall put forward an alternate approved product for review by the hydraulic consultant.

10.5 Testing

Refer to "Completion" section for details.

Provide a hydrostatic or air test to choke level for a minimum period of 15 minutes. Complete certification forms for each test carried out and submit to the Hydraulic Consultant.

10.6 Trade Waste Plumbing and Drainage

Supply and install all trade waste plumbing and drainage from fixtures to the sewer connection indicated on drawings. Provide all necessary pipes, stacks, vents, junctions, bends, expansion joins, floor wastes, excavation, supports, backfilling, testing, plant and all sundry equipment required for the installation.

Co-ordination with all other disciplines shall be carried out prior to any setting out, excavation and pipe installation taking place.

Execute the works, using only materials and structures as approved by the local Authority.

Supply and install inspection openings as indicated on drawings. Inspection openings shall incorporate a stainless-steel trap screw when installed external of the building and chrome plated finish when installed internally and of the clamp down type where in vinyl areas.

Where installed internally all drainage services shall be installed in a position to maintain maximum available head height. Note: A maximum head height to basement and car parking levels shall be maintained in accordance with the authority (local council) and Australian Standard requirements.

10.7 Invert Levels

Liaise with the civil contractor, utility provider and where required conduct a survey to confirm the depth, position and suitability of the sewer drainage connection point before commencing any work and ensure that the new trade waste drainage can gravitate to the proposed connection point.

All proposed invert levels on the drawings are recommended only and are based on information available at the time of design. Invert levels must be checked on site before excavation or installation of pipe work to ensure drainage can gravitate to the relevant connection points and correct cover and gradient can be achieved.

The Superintendent shall be notified immediately of any changes required in order to complete the work.
Redundant work resulting from failure to comply with this requirement will void any claims made.

No work is to commence until this has been completed.

NOTE:
In accordance with AS 3500.2:

- Where an 'equal 45° junction' is used to make a connection to a branch drainage line, the entry level of the branch drain shall be rotated at the junction with an incline of not less than 15° above the horizontal. The invert of the branch drain shall at least 70mm higher than graded drain to which it connects.

- Where an 'unequal sized 45° junction' is used to make a connection to a branch drainage line, the invert of the branch drain shall be rotated at the junction to ensure the invert of the unequal branch is at least 10mm higher than the soffit of the drain to which it connects.

10.8 Vents

Terminate vents through the roof at a minimum height of 150mm or as per AS 3500.2.2. requirements. Vent pipes above roof(s) shall be PVC-U DWV grade. Provide approved type vent cowls, of material to match the vent pipes.

Each vent penetration shall be flashed with a "Dektite" made of a flexible and UV resistant EPDM, as manufactured by Deks and shall be installed according to manufacturer's instructions.

The vent shall be painted to match the roof colour where require. (refer to Architects documentation)

10.8.1 Air Admittance Valves

Where indicated on the drawings, Air Admittance Valves (AAV's) complying with requirements of AS3500.2 and AS4936 may be only be utilised where approved by the local authority for Trade Waste Venting.

They shall not be used for upstream venting of a main drain, used as stacks vents where greater than 10 levels, nor installed as RVA (Reduced Velocity Stack) stack vents.

All AAV's shall be equal to Studor and shall be installed in a readily accessible location, allowing for servicing and or replacement.

10.9 Sanitary Fixtures

Refer to architectural documentation for the sanitary fixture selection.

Allow all costs to install all sanitary fixtures to the manufacturer's instructions ensuring that the installation is complete with all necessary supports, fixings and the like. All fixtures are to be installed securely and neatly into position.

Coordinate with other trades to provide cut outs in bench tops and hole preparation as required. Allow to provide technical details and samples of fixtures to enable others to perform cut-outs.

All Fixtures, appliances and brackets shall be first quality.

All exposed brackets shall be white enamelled, unless noted otherwise.

All exposed connections shall be chrome plated unless specified in white PVC or otherwise.

Obtain a written guarantee from the manufacturer for a period of two years. The guarantee shall cover all costs associated with replacement of a faulty article within a two year period.

10.10 Fixture Traps

Fixture traps shall be universal traps of approved manufacture and of equal material to which their discharge connects.

Chrome plated copper where basins or sinks are exposed

Traps shall be 50 mm diameter for sinks.

10.11 Reducing fittings

All reducing fittings installed within the drainage system are to be eccentric taper fittings / Level Invert Taper (LIT) type and shall be installed obvert to obvert to allow maximum air movement.

10.12 Inspection Openings & Expansion Joints

Expansion joints & inspection openings are to be located on all new services to the requirements of local authorities and manufactures recommendations.

Additional openings shall be provided to facilitate flushing of whole system and for testing purposes as required.

All inspection openings shall be located such that they are readily accessible and where located externally are provided with in-ground expansion joints to suit the expected ground movement.

HDPE is a reactive material with thermal expansion or contraction caused through temperature differences, with an average linear expansion coefficient of 0.2mm/m°C. The contractor shall allow to install sufficient expansion joints and suitable ‘sliding brackets’ as required, where in doubt contact the manufacture for details

10.13 Concrete Support Pads

Where drainage services are located in-ground, ensure to provide a minimum 100mm thick concrete (25mpa) support pad located below all sanitary and trade waste drainage turn-ups, gullies, floor wastes, clear outs, access openings, inspection openings.

All concrete pads are to be visible for inspection prior to backfilling.

10.14 Floor Waste Gullies

Floor wastes shall be 100mm PVC-U trap of self-cleansing pattern as indicated on the drawings.

For all floor wastes located within tiled area provide a water proof flanged “Wondercap” within the riser to avoid cement droppings from entering the trap.

Generally, floor waste gullies shall be of type, which matches the floor finish as selected by the architect. (eg Vinyl, Tile, Concrete, Timber finish etc) The grate and surround shall be chrome plated in finished areas.

Floor waste gullies shall be located at the lowest point of the floor within the room. The Sub-contractor shall ensure that the grate and surround are flush with the floor in order to facilitate the effective drainage of spills.

Where located in-ground, ensure to provide a minimum 100mm thick concrete (25mpa) support pad located below.

Where commercial laundry tubs and/or commercial washing machines discharge to a floor waste, ensure to provide anti-foaming devices within FWG’s.

Refer to architectural documentation for the sanitary fixture selection.

10.15 Bucket Trap Floor Wastes

Bucket trap floor wastes shall be 100mm HDPE trap of self-cleansing pattern as indicated on attached drawings, and shall match the floor finish as selected by the architect. (eg Vinyl, Tile, Concrete, Timber finish etc)

The internal dry basket arrestor, shall be equal to “Eco-guardian” or as approved by the local authority.

For all floor wastes located within tiled areas provide a water proof flanged “Wondercap” within the riser to avoid cement droppings from entering the trap. Wondercaps are available by phoning (07) 5539 3665.

10.16 Grease Arrestor

Supply & install in the position as indicated on the attached drawings an Halgan grease arrestor. Install in accordance with the manufacturer’s requirements complete with venting to atmosphere.

Provide and install a multi-part sectional gas and airtight Class D lids (unless noted otherwise), covers, frames and risers to the top of the arrestor to match the finished surface levels.

Where installed in-ground, ensure to allow for and coordinate the installation of suitably sized and load rated concrete grease trap slab over, to suit the manufactures requirements.

Where risers are installed to suit FSL’s, allow to extend the internal 100mm riser complete with a screw cap to within 200mm of the underside of the access lid. (This is to allow for cleaning & maintenance of the internal sludge control device ‘SCD’ fitted within the Grease Trap)

Ensure to comply with the local authorities’ anomalies, such as the addition of extra COS points, trapped outlets or manholes of a particular size for sampling where required.

10.17 Plug and wastes

All plug & wastes installed within sinks and basins shall be of brass chrome plated construction, unless noted otherwise.

10.18 Tundishes

Supply and install Chrome Plated copper tundishes made from 1.2mm sheet copper and finished with rolled edges. Tundishes shall be connected to a threaded outlet of 20mm unless indicated otherwise on drawings and fitted to universal "P" or "S" traps where indicated.

Chrome plate where positioned in finished areas.

Ensure that tundishes are installed in a visible and accessible location with physical air gaps as stipulated in AS/NZS 3500.2

10.19 Acoustic Requirements

Where trade waste plumbing and drainage is located within any habitable and non-Habitable spaces acoustic rated insulation shall be installed to meet NCC requirements.

Refer to “Materials” acoustic lagging section for details.

Note: Where an acoustic report has been provided, the hydraulic contractor shall comply with the acoustic performance requirements stipulated in the acoustic report.

10.20 Reactive Soil & Expansion Joints

Where drainage is being installed within reactive soils (Class M, H1, H2 & E) the hydraulic services sub-contractor shall install swivel joints and expansion joints in accordance with the structural engineers’ recommendations & in accordance with the Hydraulic Details.

All swivel joints and expansion joints shall be ‘Plastec’ or approved equal, and match or exceed the Ys value and shall be set at the mid-way point.

10.21 CCTV Survey & Jetting

On completion of the installation of all services and after Jetting / flushing and clearing of debris and before application for Certificate of Practical Completion, allow to undertake a CCTV survey and video recordings of all gravity drainage systems. The survey is to utilise the ‘As-built drawings’ as a reference to each section that has been surveyed.

Where deflects, blockages or other issues are noted within the recordings the contractor shall rectify.

Include all recordings in the O&M manual, on suitable USB drive.

11. STORMWATER AND DOWNPIPES

11.1 General

Comply with the General Hydraulic Services Requirements. The system includes for the supply, install, test and commission the stormwater drainage services to the requirements of Nation Construction Code of Australia, Plumbing Code of Australia, AS/NZS 3500.3, relevant Codes, the local Authority, utility provider and to the satisfaction of Superintendent and hydraulic consultant.

The work in this part comprises the supply and installation including the connection of sub-soil drainage, pipework, clear-outs, fittings, junctions, labelling, downpipes together with such minor works as necessary to form complete and satisfactory systems.

11.2 Inspections

Give sufficient notice so that inspections may be made at the following stages:

- Excavated surfaces
- Installation of pipework
- Testing of installation
- Cleaning of pipework at completion

11.3 Pipework and Materials

Refer to "*Materials*" section for details.

All materials used shall comply with SAA MP52 and be water marked, unless otherwise required by the statutory authority.

11.4 Testing

Provide a hydrostatic or air test to choke level for a minimum period of 15 minutes. Complete certification forms for each test carried out and submit to the Hydraulic Consultant.
Refer to "*Completion*" section for details.

11.5 Stormwater Drainage

The extent of the stormwater works is to provide stormwater connections to all downpipes, grated drains, sumps, planter box outlets and discharge points as shown on attached drawings.

Allow for the entire stormwater system and supply and install all necessary pipes, junctions, bends, inspection openings, excavation and sundry equipment to convey stormwater to the discharge points.

The locations of pipes indicated on the drawings are diagrammatic. Pipeline positions shall be determined on site in conjunction with all other disciplines to ensure adequate co-ordination of all services and elements. Co-ordination shall be carried out prior to any setting out, excavation and pipe installation takes place.

The entire installation shall provide adequate drainage for the project and be to the approval of the Local Authority.

Pipelines shall be laid true to line and bore from point to point.
Unless otherwise indicated on drawings, pipelines shall be graded to a minimum of 1% - 1:100.

11.6 Downpipes

Downpipes are to be located in dedicated services risers within the building cores wherever possible, however they can be cast into the structure if approved by the Structural Engineer and shall typically be installed in HDPE where cast-in. They can also be noted as being fixed externally of the building. Refer to hydraulic & architectural drawings for exact details.

The installation of downpipes shall be carried out by the Hydraulic Sub-contractor.

Downpipes shall be installed straight and parallel to the structure and the works carried out in a neat tradesman like manner. Where overflow drainage is documented, allow to install as noted within the downpipe section.

Downpipe material shall be as previously specified "*Materials*" sections and be accordance with BAC requirements

11.7 Acoustic Requirements

Where storm water drainage is located within any habitable and non-habitable spaces acoustic rated insulation shall be installed to meet NCC requirements.

Refer to "Materials" acoustic lagging section for details.

Note: Where an acoustic report has been provided, the hydraulic contractor shall comply with the acoustic performance requirements stipulated in the acoustic report.

11.8 Gutters

The installation of gutters and associated sumps shall be carried out by the roofing or plumbing contractor (depending on final contractual arrangements) - Confirm arrangement with Builder.

Allow to coordinate the installation of downpipes with the proposed gutter sumps or outlets. Seal all connections to the gutter outlets provided.

Allow to provide expansion joints to box gutters, in accordance with AS 3500 part 3 complete with minimum 50 mm expansion space and suitable saddle flashing. Refer to Architectural & BAC documents for box gutter materials.

11.9 Fire Test Drains

The installation of fire test drains shall be carried out by the Hydraulic Sub-contractor. where drainage points are required for sprinklers, provide capped points for continuation for the sprinkler trade, coordinate final locations on-site.

Fire test drains shall be installed in medium duty galvanised steel, matching that of the hydrant system pipework system complete victaulic couplings.

Allow all costs to transition medium duty galvanised steel pipework to PVC and connect into the stormwater system as noted on the drawings.

Ensure a threaded connection suitable to the local fire brigade hydrant hose is provided, complete with cap and chain.

11.10 Rain Water Outlet

Supply and install approved, flat or domed type rain water outlets cast in the concrete structure in positions indicated on the drawings.

Set grates, flush and matching with adjoining surfaces. Grates shall have integral puddle flange and membrane clamp.

Rainwater outlets shall match the design flow rates, with a maximum of 30 mm of calculated head over shall not be exceeded to suit the required flow rate.

- Where <u>vehicle traffic</u> maybe expected the grates shall be load rated to match the expected vehicular load.
- Where <u>public foot traffic</u> is expected, high-heel friendly grate patterns shall be installed.

All Rainwater outlets shall be equal to "*SPS*" Truflo RWO outlets (Speciality Plumbing Supplies)

Note: Balcony type outlets are <u>not permitted to</u> be used for RWO (Rain Water Outlets). Balcony outlets & shower outlets do not provide sufficient flow rates.

Note: All Roof Terraces & Parapet Roof structures shall be provided with a suitable means of overflow.

11.11 Planter Drain Outlets

Supply and install approved, planter drain outlets cast in the concrete structure in the approximate positions indicated on the drawings.

Set grate bodies flush with the internal surface level of the planter bed and they shall have integral puddle flanges and membrane clamp.

Surface sumps shall be installed to suit the final surface level.

- Ensure to position outlets and pipework, with sufficient room to allow water proofing to be installed in around risers.

Planter drain outlets shall be equal to “SPS” (Speciality Plumbing Supplies)

11.12 Trench Grates & Sumps

All Trench Grates shall be pre-fabricated modular items complete with matching lock down grates and shall be equal “*ACO*” and installed in accordance with manufactures requirements and as noted on the drawings.

They shall be concrete encased to the requirements as noted by the manufacture with grates set, flush and matching with adjoining finished surfaces.

All trench grates shall be a minimum of Class ‘D’ unless noted otherwise on the drawings.

Where general public foot traffic is expected, High Heel & Slip Resistant grate patterns shall be installed. Trench grates shall match the expected site flows.

Ensure trench grates are cleaned and free of debris prior to completion and handover.

Note: Where Sumps are required, they shall meet DoE requirements complete with High Heel & Slip, Finger & Toe Resistant grates. They shall also be provided with secure fastenings to grates.

11.13 Reactive Soil & Expansion Joints

Where drainage & downpipes are being installed within reactive soils (Class M, H1, H2 & E) the hydraulic services sub-contractor shall install swivel joints and expansion joints in accordance with the structural engineers’ recommendations & in accordance with the Hydraulic Details.

All swivel joints and expansion joints shall be ‘Plastec’ or approved equal, and match or exceed the Ys value and shall be set at the mid-way point.

11.14 CCTV Survey & Jetting

On completion of the installation of all services and after Jetting / flushing and clearing of debris and before application for Certificate of Practical Completion, allow to undertake a CCTV survey and video recordings of all gravity drainage systems. The survey is to utilise the ‘As-built drawings’ as a reference to each section that has been surveyed.

Where deflects, blockages or other issues are noted within the recordings the contractor shall rectify.

Include all recordings in the O&M manual, on suitable USB drive.

12. COLD WATER SUPPLY

12.1 General

Comply with the General Hydraulic Services Requirements. The system includes for the supply, install, test and commissioning of services to the requirements of Nation Construction Code of Australia, Plumbing Code of Australia, AS/NZS 3500.1, relevant Codes, the local Authority, utility provider and to the satisfaction of Superintendent and hydraulic consultant.

The work in this part comprises the supply and installation including for all pipework, valves, backflow, bends, offsets, brackets, metering, pumps, tapware / faucets labelling and sundry equipment together with such minor works as necessary to form complete and satisfactory systems. Carry out all shoring and backfilling as required.

Provide the fittings necessary for the proper functioning of the water supply system, including taps, valves, backflow prevention devices, pressure and temperature control devices & strainers.

12.2 Inspections

Give sufficient notice so that inspections may be made at the following stages:

- Excavated surfaces
- Installation of pipework
- Testing of installation

12.3 Pipework and Materials

Refer to “*Materials*” section for details.

All materials used shall comply with SAA MP52 and be water marked, unless otherwise required by the statutory authority.

Pipelines used in rough-ins shall be installed in straight lines and offset using right angle manufactured bends. Offsets shall be manufactured around other piping and installed in a tradesman like manner. Pipework shall be securely fastened using approved saddle clips.

All pipes installed external, shall be UV resistant or protected from UV with cladding or similar protection.

12.4 Sizing

Cold water services shall be installed in compliance with AS 3500.1. Minimum pipe sizes shall be 20 mm diameter. Exception is 15 mm diameter may be used for a single fixture for a maximum distance of 6 metres.

12.5 Testing

Complete certification forms for each test carried out and submit to the hydraulic consultant. Testing certificates shall be included within the Operation and Maintenance Manuals.

Refer to “Completion” section for details.

12.6 Connection to Fixtures

Each fixture connection shall be made using a threaded 19 BP screwed to a timber noggin to prevent any movement of the fittings.

Provide at each fixture a flexible connector equal to ‘Abey – Polyamide Hi Class Hooker’ that incorporates an anti-kinking design. These connections shall allow removal and replacement without the need to adjust the connections.

Provide a chrome plated cover plates at each connection neatly secured to the wall, complete with mini stop valve and flexible connector.

Refer to “*Materials*” section for details.

12.7 Control Valves

Supply and install control valves to each group of fixtures. Locate behind access panels or within accessible ducts.

Ensure valves are labelled, refer to “*Materials*” section for details.

12.8 Pressure Reduction

Provide pressure reduction valves on individual branch lines to ands as necessary to maintain pressure limits to a maximum of 350kPa to each fixture

12.9 Mini Stop Valves

At each wall connection for both hot and cold water to all basin and sink mixers, supply and install chrome plated 15 mm right angled, chrome plated mini stop valves.

12.10 Fittings and Accessories

Provide all fittings necessary for the proper functioning of the water supply system, including taps, valves, backflow prevention devices, water hammer devices, pressure and temperature control devices & strainers.

12.11 Water Meters

The location, size, selection and installation of all authority water meters shall be in accordance with the relevant Authority requirements.

Generally, local potable cold water meters are required at the following locations:-

- Main incoming feed supply.
- Each tenancy.

Approved sub-meters shall be installed as noted on the drawings and shall meet the requirements of BAC including tagging, minimum and maximum heights, clearances, drainage and bunding of meter cupboards

All meters are to have a reed switch, which permits connection to an Automatic Meter Reader (AMR) and Building Management System (BMS) and suit BAC requirements

12.12 Hose Taps

Each hose tap shall be screwed into an "elbow with backplate", each elbow (backplate) to be securely fixed with 3 no. Stainless steel screws.

Hose taps to be generally 450 mm above floor.

External - Hose taps to have control valves and be fitted with a hose connector vacuum breaker (HCVB), RMC or approved equal. Shear grub screw off prior to practical completion – only after fully flushing the water service.

- External hose taps shall be vandal proof with removable key.
  Allow to provide removable vandal keys equal to the number of external hose taps.

Provide hose taps to each external landscaped area and planter area.
Provide hose taps for garbage and laundry areas.

Internal - Hose taps shall be 15 mm C.P. finish to match other faucets.

12.13 Flow Restriction

All fixtures shall meet the below noted maximum flow rates & restrictions.

Where fixtures do not meet the below noted requirements, supply and install flow control devices. The devices shall be equal to "CONSERV" Pty Ltd.

- Shower 9 L/M
- Basin 6 L/M
- Sink 6 L/M
- Toilet cistern 4.5 FULL FLUSH / 3 HALF FLUSH

12.14 Mini Stop Valves

At each wall connection for both hot and cold water to all basin and sink mixers, supply and install chrome plated 15 mm right angled, chrome plated mini stop valves.

12.15 Water Hammer

Ensure that water piping and fittings are so installed as to prevent water hammer occurring within the system. Water hammer is a defect and should water hammer occur, the Sub-Contractor is responsible for the correction and rectification of the problem to the satisfaction of the Superintendent.

The Sub-Contractor shall take all precautions to prevent water hammer. These may include:-

- Provide and install water hammer arrestors in appropriate locations to reduce water hammer.
- Install additional pipe clamps as required to reduce water hammer.
- Make adjustments to pipe locations as required to reduce water hammer.
- Install pressure regulating valves as required to lower pressure.

12.16 Thrust Blocks

To all changes of direction on unrestrained jointed pipelines below ground install concrete thrust blocks to restrain the internal operating pressures of the pipeline under all conditions. Concrete mass shall be poured around and behind fittings and bear against virgin soil material. A minimum of 0.75 cubic metres of concrete shall be used at each position.

Refer to “Installation” section for more details.

12.17 Pipework in Roof Spaces

Provide thermal insulation to cold water piping where installed within unventilated roof spaces. Insulation shall be equal to thermal insulation for hot water flow and return piping as indicated in this specification.

13. HOT & WARM WATER SUPPLY

13.1 General

Comply with the General Hydraulic Services Requirements. The system includes for the supply, install, test and commissioning of services to the requirements of Nation Construction Code of Australia, Plumbing Code of Australia, AS/NZS 3500.4, relevant Codes, the local Authority, utility provider and to the satisfaction of Superintendent and hydraulic consultant.

The work in this part comprises the supply and installation including for all hot water pipework, valves, thermal insulation, backflow, bends, offsets, brackets, metering, pumps, tapware / faucets, hot water units and plants, labelling, and sundry equipment together with such minor works as necessary to form complete and satisfactory systems.

Provide the fittings necessary for the proper functioning of the hot water supply system, including taps, valves, backflow prevention devices, pressure and temperature control devices & strainers.

13.2 Inspections

Give sufficient notice so that inspections may be made at the following stages:

- Excavated surfaces
- Installation of pipework
- Testing of installation

13.3 Pipework and Materials

Refer to "*Materials*" section for details.

All materials used shall comply with SAA MP52 and be water marked, unless otherwise required by the statutory authority.

Pipelines used in rough-ins shall be installed in straight lines and offset using right angle manufactured bends. Offsets shall be manufactured around other piping and installed in a tradesman like manner. Pipework shall be securely fastened using approved saddle clips.

All pipes installed external, shall be UV resistant or protected from UV with cladding or similar protection.

13.4 Testing

Complete certification forms for each test carried out and submit to the hydraulic consultant. Testing certificates shall be included within the Operation and Maintenance Manuals.

Refer to "Completion" section for details.

13.5 Connection to Fixtures

Each fixture connection shall be made using a threaded 19 BP screwed to a timber noggin to prevent any movement of the fittings.

Provide at each fixture a flexible connector equal to 'Abey – Polyamide Hi Class Hooker' that incorporates an anti-kinking design. These connections shall allow removal and replacement without the need to adjust the connections.

Provide a chrome plated cover plates at each connection neatly secured to the wall, complete with mini stop valve and flexible connector.

Refer to "*Materials*" section for details.

13.6 Control Valves

Supply and install control valves to each group of fixtures. Locate behind access panels or within accessible ducts.

Ensure valves are labelled, refer to "*Materials*" section for details.

13.7 Pressure Reduction

Provide pressure reduction valves on individual branch lines to ands as necessary to maintain pressure limits to a maximum of 350kPa to each fixture

13.8 Mini Stop Valves

At each wall connection for both hot and cold water to all basin and sink mixers, supply and install chrome plated 15 mm right angled, chrome plated mini stop valves.

13.9 Fittings and Accessories

Provide all fittings necessary for the proper functioning of the water supply system, including taps, valves, backflow prevention devices, water hammer devices, pressure and temperature control devices & strainers.

13.10 Flow Restriction

All fixtures shall meet the below noted maximum flow rates & restrictions.

Where fixtures do not meet the below noted requirements, supply and install flow control devices. The devices shall be equal to "CONSERV" Pty Ltd.

- Shower 9 L/M
- Basin 6 L/M
- Sink 6 L/M

13.11 Thermostatic Mixing Valves

Supply and install a thermostatic mixing valve (TMV) to control the temperature of the warm water to the outlet points as shown. Thermostatic mixing valves shall be installed within in-wall SS boxes or as indicated on the attached drawings.

The Thermostatic mixing valve shall be complete with control valves, in-line strainers and unions.

The entire installation shall conform to the local authorities' requirements, and be in accordance with the Code of Practice for Thermostatic Mixing Valves.

Provide all work for the installation, commissioning instructions, service and maintenance instructions, copies of commissioning test reports and valve warranties or guarantees as required by the Code of Practice.

The entire installation shall be left in a "User-Safe" condition and the issuing of the necessary installation and commissioning certificates shall be mandatory prior to acceptance of the valve installation.

Valves shall be installed strictly in accordance with the manufacturer's printed instructions and ensure correct adjustment to the final installation to provide an adequate and safe supply of tepid water. Set temperature range as directed at site or as noted below.

After installation, purging of lines, reinstallation of the valve and commissioning, the temperature of the valves shall be set to 45°C to all required areas.

The TMV's shall be a pressure balanced TMV, equal to 'F.M Mattsson' and contain built-in 'thermal flush capability' to allow hot water disinfection through the outlets and valve.

Refer to "*Materials*" section for details.

13.12 Expansion and Contraction

Make adequate provision for expansion and contraction by the provision of expansion material so that under all working conditions no strain is imposed on pipework or fittings.

Pipes located in walls shall be provided with sufficient insulation so that expansion and contraction does not impose a strain on the pipework or finished surfaces.

No pipes will be allowed within or under concrete slabs where slab is on ground. Pipes installed within concrete slabs shall be installed within a preformed conduit which allows removal of the pipe.

Branch pipes off straight lengths of unrestrained supply pipes shall incorporate a minimum of 3 long radius pipe bends and a straight length of pipe not less than 1000mm long between the branch connection and the first branch pipe restraint to facilitate movement of the supply pipe without imposing strain on the pipe connections and fittings.

13.13 Hot Water Heaters

Hot water heaters shall be fully mains pressure type complete with all necessary valves in accordance with AS 3500.4 and Local Authority requirements, complete with drainage and safe trays necessary to complete the installation. The Sub-contractor shall allow for the supply, installation and commission all of the heaters. The hot water heaters shall be located as indicated on the drawings.

Each hot water heater shall be factory pressure tested and provided with Corrosion protection (sacrificial anodes) where required.

Where multiple heaters are installed in banks use the manufacturer's standard manifold arrangement to provide equal flow thorough each heater in the bank.

The temperature and pressure relief valves (TPRV) shall conform with the requirements of AS 1357 and extend to drainage in an approved manner. Allowance for cold water expansion control valves (ECV) shall be made and installed as required by local authorities, with discharge over drainage complete with air gap.
Where heat pumps are installed, allow to extend condensate discharge to drainage point.

Electric hot water heaters shall be in accordance with AS 1056, Section J of the BCA and be tested and approved by the electrical supply authority.

All hot water plants requiring primary circuit heat transfer pumps, shall have the pumps supplied by the hot water heater manufacturer.

All vented under sink hotwater units shall be supplied complete with Isolation valve and Pressure Limiting Valve set at 350 kPa. And shall be supplied with matching vented tapware.

On completion of installation, engage the water heater manufacturer to commission and test the installation. Manufacturer is to provide an installation clearance certificate.

Note:
Electronic timers shall be installed to shut down hotwater systems and boiling water units afterhours.
The shut down

Refer to “Equipment and Pumps” section for details.

13.14 Hot Water Circulation Pumps

Supply and install dual hot water circulating pumps to maintain a constant operating temperature in the hot water flow and return piping.

Pumps shall be capable of operating continuously through a 24 hour time clock with seven (7) days spring reserve, and be provided with BMS connection.

Mount pumps on wall bracket with neoprene rubber mounting.

Refer to “Equipment and Pumps” section for details.

13.15 Air Elimination Device

Where a hotwater circulation system is installed, Air Elimination Valves shall be installed.

Air elimination valves shall be installed:

- At the highest point within the hotwater circuit. (typically)
- In an accessible location.
- Complete with an individual isolation valve.
- With a connection to drain, with discharge being visible and over a tundish.
- Installed in accordance with the manufacture’s requirements

Ensure air elimination valves exceed or match the following requirements:

- Contain watermark approval, for use on potable water services.
- Working pressure = 1400kPa
- Operating temperature = 200°C

13.16 Hot Water Circulation System Diagram

For all class 2 to 9 buildings & where a hotwater circulation system is installed, allow all costs to supply and install a water, fade and weather-resistant diagram, which shall be installed, adjacent to the circulation pumps.

The diagram shall meet AS 3500.4 – section 9.5 requirements.

13.17 Temperature Gauges

Temperature gauges shall be Ambit ME-PF-100-120 or equal complete with temperature well. Install temperature gauges on the outlets of water heaters, on the flow and return lines and adjacent to all Balancing valves. Ensure all Temperature gauges are provided with an individual isolation valve

- It is the contractor's responsibility to ensure installed temperature gauges read true and match the design temperatures of the system. The contractor will be required to reinstall incorrectly installed temperature gauges at their own expense.

14. RAIN WATER SUPPLY

14.1 General

Comply with the General Hydraulic Services Requirements. The system includes for the supply, install, test and commissioning of services to the requirements of Nation Construction Code of Australia, Plumbing Code of Australia, AS/NZS 3500.1, relevant Codes, the local Authority, utility provider and to the satisfaction of Superintendent and hydraulic consultant.

The work in this part comprises the supply and installation including for all rain water pipework, valves, backflow, bends, offsets, brackets, metering, pumps, filtration, solenoid valving, tapware / faucets, hot water plants and sundry equipment together with such minor works as necessary to form complete and satisfactory systems. Carry out all shoring and backfilling as required.

Supply and install all rain water pipes from the rain water storage tanks to all fixtures, fittings and faucets requiring rainwater reticulation, refer to hydraulic drawings for details (Typically – WC's, UR's and irrigation points)

Provide the fittings necessary for the proper functioning of the rainwater supply system, including taps, valves, backflow prevention devices, pressure control devices, strainers and rainwater pressure pumps.

- Ensure clear labelling is provided as required by the local authority and in accordance with AS 1345.

14.2 Inspections

Give sufficient notice so that inspections may be made at the following stages:

- Excavated surfaces
- Installation of pipework
- Testing of installation

14.3 Pipework and Materials

Refer to "*Materials*" section for details.

All materials used shall comply with SAA MP52 and be water marked, unless otherwise required by the statutory authority.

Pipelines used in rough-ins shall be installed in straight lines and offset using right angle manufactured bends. Offsets shall be manufactured around other piping and installed in a tradesman like manner. Pipework shall be securely fastened using approved saddle clips.

All pipes installed externally shall be UV resistant or protected from UV with cladding or similar protection.

14.4 Testing

Provide a water pressure test of 1500Kpa for a minimum period of 30 minutes. Disconnect any equipment connect to the service not rated to the test pressure, before testing commences.

Complete certification forms for each test carried out and submit to the hydraulic consultant. Testing certificates shall be included within the Operation and Maintenance Manuals.

Refer to "Completion" section for details.

14.5 Connection to Fixtures

Each fixture connection shall be made using a threaded 19 BP screwed to a timber noggin to prevent any movement of the fittings.

Provide at each fixture a flexible connector equal to 'Abey – Polyamide Hi Class Hooker' that incorporates an anti-kinking design. These connections shall allow removal and replacement without the need to adjust the connections.

Provide a chrome plated cover plates at each connection neatly secured to the wall, complete with mini stop valve and flexible connector.

Refer to "*Materials*" section for details.

14.6 Control Valves

Supply and install control valves to each group of fixtures. Locate behind access panels or within accessible ducts.

Ensure valves are labelled, refer to "*Materials*" section for details.

14.7 Pressure Reduction

Provide pressure reduction valves on individual branch lines to ands as necessary to maintain pressure limits to a maximum of 350kPa to each fixture

14.8 Mini Stop Valves

At each wall connection, supply and install chrome plated 15 mm right angled, chrome plated mini stop valves. Ensure signage is clearly noted on the pipework and cover plate

14.9 Fittings and Accessories

Provide all fittings necessary for the proper functioning of the water supply system, including taps, valves, backflow prevention devices, water hammer devices, pressure and temperature control devices & strainers.

14.10 Rain Water Pumps

The rain water pressure pumps shall be installed where indicated on the drawings, complete with vibration mountings.

The rain water pump system shall be installed complete with a potable water by-pass system, plumbed directly into the rain water reticulation pipework and operated via a low level alarm / float.

Commissioning report shall be included within the operation and maintenance manual.

Refer to "Equipment and Pumps" section for details.

14.11 Rain Water Harvesting Tank

Where indicated on the attached drawings supply & install rainwater harvesting tank/s complete with first flush system and overflow connecting to the stormwater drainage system.

Allow to install above ground rainwater tank/s in accordance with the tank manufacturers' requirements for support.

At the base of above ground tanks, install a capped 50mm ball valve to allow the contents of the tank to be drained. Connect the tank drain to the overflow drain.

Allow to install rainwater tank/s in accordance with the tank manufacturers' requirements.

All connections are to be mosquito proof in accordance with QDC requirements.

Supply and install a pressure pump system, complete with control panel, mains water by-pass, backflow protection and extend rainwater service to serve all toilets and urinals as part of the base build works

Refer to "Equipment and Pumps" section for details.

15. FIRE SERVICES

15.1 General

Comply with the General Hydraulic Services Requirements. The system includes for the supply, install, test and commissioning of services to the requirements of Nation Construction Code of Australia, Plumbing Code of Australia, AS 2491, AS 2441, AS 1851, relevant Codes, the local Authority utility provider, the local Fire Brigade (QFD & ARFF) and to the satisfaction of Superintendent and hydraulic consultant.

Allow for the installation of all hydrants, fire hose reels and hydrant booster assembly, cabinet and bollard protection, from local authority's water main as indicated on the attached drawings.

Include for all piping, thrust blocks, fittings, valves, hydrant valves, control equipment, labelling, and other sundry items of equipment as required for the installation in accordance with the Building Code of Australia, AS 2419 and AS 1221.

Fire hose reels are not to be used as a water supply source during construction. All damaged fire hose reels shall be replaced without incurring additional costs to the contract.

15.2 Inspections

Give sufficient notice so that inspections may be made at the following minimum stages:

- Excavated surfaces
- Installation of pipework
- Performance testing of installation
- Fire Brigade Testing of installation

15.3 Pipework and Materials

Refer to "*Materials*" & "*Installation*" section for details.

Use equipment authorised and approved for Fire Protection use only.

Ensure thrust blocks are utilised in-ground at all changes of direction.

15.4 Testing

Provide a water pressure test (hydrostatic) to the greater of 1700Kpa in accordance with 'Tests 1 to 4' in appendix S of AS 2419.1-2021.

- On completion and prior to the local fire brigade being contacted to undertake the final inspection. Allow all costs to engage an appropriate company to carry out performance flow and pressure tests of the installed fire system. Record pressure and flow test results in the approved manner and advise outcome in writing.

Refer to "Completion" section for details.

15.5 Fire Hydrant Landing Valve

Landing Valve shall be:

a) Be installed in accordance with AS 2419 - Fire Hydrant Installations.
b) Approved by the local water supply authority.
c) Approved by the local Fire Brigade (QFD & ARFF)
d) Of all bronze construction.
e) Fitted with bronze cap and chain to the outlet thread.
f) Of 65 mm diameter.
g) Installed at a height of 750 mm from centre of valve to ground level to maximum of 1200mm.
h) Identified by metal or ultraviolet (UV) resistant disc of not less than 20mm diameter displaying the fire hydrant number as shown on the block plan.

- External Hydrants shall consist of a dual hydrant landing valve arrangement. Where mounted in grassed or landscaped areas, install each external hydrant complete with a 450 x 450 x 100 thick concrete base finished 25mm above ground level.

Notes:
Where hydrants are prone to vandalism, ensure to provide anti-vandal custom made lock-able hydrant valve handle covers. Ensure to provide 003 keyed matching locks.

Where external hydrants are located where possible vehicle impact damage may occur, allow to provide sufficient bollard protection to meet local fire brigade requirements & BAC requirements

15.6 Fire Hose Reel

Fire Hose Reels shall be:

a) Be installed in accordance with AS 2441.
b) 36m length and be equal to Galvin Engineering “Red Emperor”
c) Approved by the local water supply authority.
d) Approved by the local Fire Brigade.
e) Supplied with a 25 mm diameter isolation valve.
f) Be mounted with the centre of the hub 1500 mm above floor level.
g) Be installed downstream of an approved double check valve. If installed adjacent to a high hazard area, this device shall be upgraded to the RPZD.
h) Where free standing, ensure to provide matching and suitable bolt down support post & matching lockable cabinet.
i) Where installed externally or as noted they shall be provided within a suitable matching and lockable powder-coated red wall or post mounted fire hose reel cabinet.

Secure reel hub to wall by means of 10mm diameter 'Dynabolt' type fittings to masonry walls or by other means of approved fixings, in accordance with manufactures requirements.

Note: Allow to coordinate additional suitable support framing behind Fire Hose Reels with builder, where installed on to internal walls.

Additional operating instruction sheet shall be procured and secured in a position adjacent to the reel in a permanently upright position.

Fire hose reels are not to be used as a water supply source during construction. All damaged fire hose reels shall be replaced without incurring additional costs to the contract.

15.7 Fire Booster Assembly

The Fire booster Assembly shall in accordance with AS 2419 and local fire brigade requirements. (QFD & ARFF)

The booster assembly shall be complete with compliant booster cabinet, concrete pad graded to ensure no ponding of water and local brigade approved block plan(s) fixed with the booster cabinet and fire pumproom & fire control room.

Where the fire booster assembly is installed within or affixed to the facade of the building and more than 20m away from principal pedestrian entrance allow to provide a VAD (Visual Alarm Device) to identify the fire brigade booster assembly to AS 2419.1-2021 Section 7.3.2.

The VAD (Visual Alarm Device) shall be activated by an alarm signal from a FDCIE system that serves.
- an automatic smoke detection system and alarm system, installed throughout the building
- a sprinkler system, installed throughout the building
- or combination of the above, provided the building is protected throughout.

Allow to provide a set of 003 keys to the client upon completion.

15.8 Isolation Valves – Fire Hydrant system

Isolation valves installed within a fire hydrant system shall meet AS 2419.1-2021 requirements and be located as follows:

- Where installed with a building and above ground, located not more than 2.4m to the valve handle above the FFL.
- Installed in a fire-isolated stair way or
- Installed in a plant, tank, pump room or
- Where fire -isolated stairs are not provided, IV’s shall be installed not more than 4m from an exit leading to a road or open space

15.9 Pressure gauges

All pressure gauges shall comply with AS 1349, they shall be a liquid-filled bourdon type

They shall be full scale reading of not less than 125% of the system hydrostatic test pressure at the point where the gauge is located. They shall installed complete with an gauge cock be of a minimum of a dimeter of 15mm.

Ensure to provide pressure gauges at the following minimum locations and in accordance with AS 2419.2021

- On the suction & delivery side of any Booster Pump
- On the delivery side of any pressure Maintenance / Jacking pump
- Adjacent to any Booster Assembly inlet connection
- At each pressure switch
- Buildings under 25m effective height – and where more then 6 hydrants are installed, pressure gauges shall be installed at the most disadvantaged hydrants
- Buildings greater than 25m effective height - at the most disadvantaged hydrants in any pressure zone & immediately upstream and downstream of any pressure reducing valve.
- Where Replay Pumps are installed pressure gauge(s) shall be installed within the booster assembly, complete with signage showing Reply Pump discharge pressures

15.10 Fire Precautions During Construction

Where required and in accordance with NCC and local authority requirements, allow all costs to provide and maintain a temporary Fire hydrant and fire hose reel protection system during the construction phase.

For assistance reference shall be made to NCC volume 1, section E1.9

15.11 Fire Hydrant Pipework and Tilt-up Panel Walls

Where Tilt-up Panel Walls are used in the construction of the development, ensure that Fire hydrant pipework is NOT mounted on, pass through, or fixed to tilt-up panel walls.

Where pipework is indicated & positioned close to or adjacent to tilt-up panel walls allow all costs to install suitable free standing pipework support ladder system, in-leu of fixing hydrant pipework to tilt-up panel walls.

15.12 Fire Rating of Pipework Supports

In accordance with AS 2419.1-2005 section 8.7.4, where fire hydrant pipework is likely to be exposed to fire in a building that is not protected by fire sprinklers, then hydrant pipework supports shall have a FRL not less than 60/-/- in accordance with AS 1530.4.

Allow all costs to supply and install fire rated pipe supports, including but not limited to; fixings, threaded rod, pipe clips and brackets, nuts, etc equal to ‘FRS’ (Fire Rated Solutions pty ltd)

Contact ‘FRS’ on mobile: 0420 622 288 or email: sales@fireratedsolutions.com.au

16. PUMPS & EQUIPMENT

16.1 General

Supply and install all control equipment necessary to operate the pumps and control equipment specified under hydraulic services

Allow for all control cabinets, mounting brackets, contactors, isolating and control switches, auxiliary switches, alarms, wiring between pump and panel, panel and level controls, connection of power supply to panel, BMS connections, electrical conduits (cast-in slab where required), step irons within chambers / tanks and other associated equipment and commissioning as necessary for the safe and effective operation of the pumps and equipment as required for the installation and in accordance with statutory requirements.

Where required, all control panels and pumps shall be intrinsically safe.

All pumps shall be mounted on a 150 mm high concrete plinth that also extends 150 mm baseplate, complete with spring mounts & shall have 75mm dia. Pressure gauges (of the bourdon type) complete with IV provided to each side of the pumps.

Fit a 'Binda' cock adjacent to all pressure gauges

16.2 Fire Hydrant Pumpset

Supply and install as shown on the hydraulic drawings a "packaged" fire hydrant pumpset complete with hydrant pump(s) and single electric jacking pump.

The entire pumpset shall be in accordance with AS 2419, AS 2941, AS 2118 & QFD & BAC requirements, including exhaust pipework to atmosphere with the following:

- 1 x Diesel pump and 1 x Electric jacking pump
- Duty: 10 l/s @ 50 m/head **(estimate provided for D&C assistance only – subject to final calcs)**
- Electrical requirements: 240v, 1.5kW & 15 A (Diesel pump)
  240v, 1.5kW & 8.9 A (Jacking pump) on a dedicated circuit

Supply and install all pipework, fittings, valves, brackets, fixings, conduits, anti-vibration joints, exhaust pipework to atmosphere, wiring and (monitoring - where required), necessary for the complete and fully operational service, as indicated on hydraulic drawings.

Note: Concrete plinths shall be provided to suit the installed fire pump with a height not less than 150 mm and shall extend 150 mm past the edge of the baseplate on all four sides of fire pump.

Ensure Control Panels are capable of BMS & FIP connection

The pump set shall be "Aline Pumps" or approved equal.
PH 1800 018 999

16.3 Hotwater Units

Supply and install as shown on the hydraulic drawings mains pressure Hot Water Units, complete with all necessary valves, concrete plinths and safe tray drainage in accordance with AS 3500 part 4 and local authority requirements.

The Hot Water Units shall be installed as noted and consist of the following:

Ensure to provide 'heat traps' to all non-circulation installations, within 1 meter from the outlet of the water heater and before the first branch.

RHEEM model No. 6N3-315 315L
- Heating element: 3 x 3.6 kW
- Safe trays and drainage shall be provided to each hot water unit
- Located: Cleaners room

16.4 Hotwater Circulation Pumpset

Supply and install as shown on hydraulic drawings a hot water circulation pump set. Pump set shall be complete with the following:

Supply and install all pipework, fittings, valves, brackets, fixings, anti-vibration joints, wiring etc, necessary for the complete and fully operational service, as indicated on hydraulic drawings.

Ensure Copper pipework is installed

- Dual hotwater circulation pumps (duty / stand-by & auto change over)
- Duty: 0.1 l/s @ 1m/h each pump **(estimate provided for D&C assistance only – subject to final calcs)**
- Electrical requirements: 240V / 10 A each pump **(estimate provided for D&C assistance only – subject to final calcs)**
- (1) Control panel, capable of BMS connection, complete with manual/on/off selector switch.
- (1) Time clock controller.
- Air Eliminator Valve with IV and drainage pipework discharging over a suitable drain point.
- Temperature gauges to the outlet of the pump set.
- Circulation hotwater diagram that is a minimum of A3 is size and water, fade & weather resistant, in accordance with AS 3500 part 4.

NOTE: Ensure circulation temperature gauges read true, where temperature gauges are found to have been installed poorly and do not provide accurate temperature readings, the contractor shall re-install at no cost to the project.

The entire pump set shall be "Aline Pumps" or approved equal.
PH 1800 018 999 - quote reference number:

16.5 Rainwater Pumps

Supply and install as shown on hydraulic drawings a "packaged" Above-ground rainwater pressure pump set, complete with main water auto-change over and backflow valve arrangement.

- Allow all costs to provide and install weatherproof "rainwater pump system enclosure" complete with hinged service doors to ensure suitable weather protection is provided to RW pumps

Dual (duty/stand-by) variable speed, constant pressure pump set

- Total duty: 1 L/s @ 35 m/h (TBC) – **(estimated provided D&C assistance only – subject to final calcs)**
- Electrical requirements: 415V, 1.5kw, 3.8 each pump - **(estimated provided D&C assistance only – subject to final calcs)**

- (1) Control panel, capable of BMS connection, complete with manual/on/off selector switch, tank / mains water operation indicator lights & tank level indicator status lights
- (1) 32mm brass solenoid valve "Normally Open" (for mains operation).
- (1) Pressure controller.
- (2) Pressure gauges (complete with IV's) to the inlet & outlet of the pumpset.
- (1) hydrostatic level senor

Supply and install all pipework, fittings, valves, brackets, fixings, anti-vibration joints, wiring, conduits etc, necessary for the complete and fully operational rainwater service, as indicated on hydraulic drawings,

The entire pump set shall be "Aline Pumps" or approved equal and shall meet project requirements
PH 1800 018 999 - quote reference number:.EST 201589

16.6 Rainwater Back-Wash Filter

Supply and install as noted on the hydraulic drawings an automatic (timer) controlled back-wash filter. It shall be designed to protect downstream fixtures & fittings from malfunctioning and damage caused by debris, sand, and foreign particles.

- Connection size: 25mm
- Filter size: 89 micron
- Power requirements: 240V GPO (24 volt)
- Manufacture: Puretec
- Model Number: IAB25

Notes:
Ensure to provide IV's either side of valve, complete barrel unions & ensure discharge is piped and directed over a suitably sized discharge point connected to drain.

Initially set the back-wash filter to operate; out of hours once (1) every, seven (7) days. Where required, adjust back-wash operation to suit on-site conditions.

16.7 Rainwater Tank(s) – Above Ground

Supply and install as shown on the hydraulic drawings above ground rainwater tanks, complete with overflows, first flush, valves, manifolding and capped drain down point and associated pipework & pipework support frames.

- 25,000 Litre tank as noted.
- 1 x 400dia Hiflow inlet strainer
- 2 x 225 high level overflows
- 25mm pump outlet
- Tank material: food-grade polymer coated – colourbond steel
- Colour: final colour to be determined
- Equal to Kingspan Water Tanks

Allow all costs to install and secure the rainwater tanks in accordance with manufacturers requirements, complete with a suitability prepared level, compacted and stable ground pad.

<u>As a Minimum</u>: Ensure the prepared ground pad is at least 300 mm wider than the diameter of the tank with allowance for sumps and associated pipework complete with a screeding of 100 mm thickness of Crusher Dust / Blue Metal (<3 mm) complete with geo-fabric base layer. Ensure that these screeding's are level, compacted and retained to prevent wash out, complete with treated pine retaining (75 mm thick) sleepers or similar.

<u>Notes:</u>
- Allow to coordinate **Final Tank Colour** with Architect - prior to ordering and installation of RW tanks.
- Allow to provide a suitable galvanised steel support frame and footings that will support the weight of the flooded inlet pipework, overflows and first flush devices.

16.8 Rainwater Tank First Flush

Supply and install as noted on the hydraulic drawings a suitable pre-rainwater tank first flush device. Allow all costs to supply complete to manufactures requirements with all required pipework connections, support brackets / framing and discharge over storm water.

- Connection size: 300mm (TBC)
- Model Number: WDCL22 - Delta commercial 225 mm (TBC)
- Chamber volume: TBC

<u>Notes:</u>
- Allow to provide a suitable galvanised steel support frame and footings that will support the weight of the flooded inlet pipework and first flush devices.

To be equal to 'Blue Mountain Co'
PH (07) 3248 9600

17. SANITARY FIXTURES & TAPWARE

17.1 General

All sanitary fixtures and taps used shall be of first quality and comply with SAA MP52 & be WaterMarked unless otherwise required by the statutory authority.

17.2 Sanitary Fixtures

Sanitary fixtures shall in accordance with the architectural schedule.

17.3 Tapware

Tapware and outlets shall in accordance with the architectural schedule.

17.4 Samples

The following minimum items shall be provided as samples for approval by the Architect prior to commence of works.

- Valve identification plate.
- In-ground Warning Tape.
- Identification Labels.
- Hot Water Insulation.
- Condensate Insulation.
- Rainwater, Balcony, Trench Grate & Sumps.
- Floor Waste Grate to suit the different floor coverings as required throughout the project.
- Bucket trap & Grate to suit the different floor coverings as required throughout the project.
- Clear Out to Surface (COS) covers for all different floor coverings as required throughout the project, including load rated cover and frames suited for the anticipated loads.
- PWD Shower hose restrictor device or bracket.
- Fixtures and Tapware as requested by the architect.